

# Manufacturing and Service Operations

## MSO 40.012

R Desmicht – Adjunct Professor SUTD

# Summary Course 3



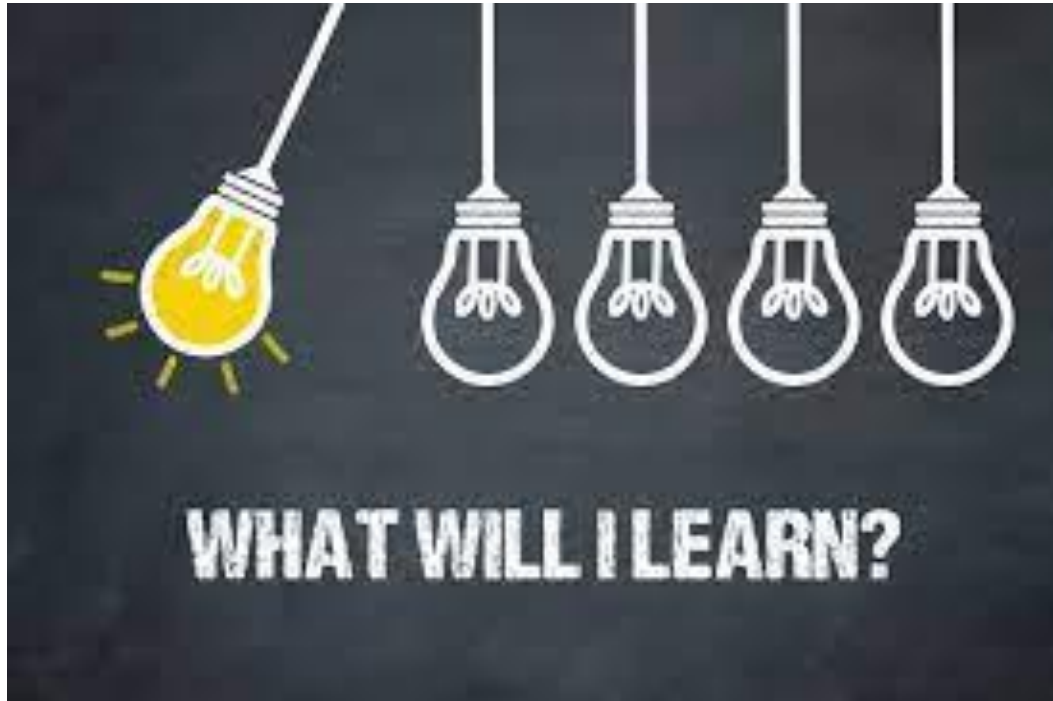
- **MPS Definition:** The Master Production Schedule (MPS) specifies the timing and quantity of specific finished goods to be produced, serving as a more detailed breakdown of the aggregate plan.
- **Planning Hierarchy:** Production planning moves from a macro-level marketing strategic plan down to mid-term Industrial Plans (S&OP), then to the MPS, and finally to short-term daily scheduling .
- **Aggregate to MPS Transition:** The process involves disaggregating product families into specific products using fixed historical ratios, current customer orders, and safety stock needs .
- **Cost Optimization:** Production planning aims to find the lowest total cost by balancing two competing factors: setup (ordering) costs and inventory holding costs .
- **Simple Heuristics:** Basic methods for solving the production horizon problem include "One Time Run" (producing everything at once) and "Lot for Lot" (producing exactly what is needed per period) .
- **Specialized Heuristics:** Advanced methods like the Silver-Meal algorithm aim to minimize the average cost per period rather than just looking at the immediate demand .

- **MRP Definition:** Material Requirements Planning (MRP) is a dependent demand technique that uses the MPS, Bill of Materials (BOM), and inventory data to determine what, when, and how much to order.
- **Bill of Materials (BOM):** This is a hierarchical listing of all components and raw materials required to manufacture a finished product .
- **Lead-Time Offsetting:** MRP uses a "time-phased explosion" to determine when to release orders for sub-components based on the final product's due date and known lead times.
- **MRP Assumptions:** Standard MRP logic assumes known demand, no capacity constraints, constant lead times, and no back-ordering.
- **MRP Limitations:** The system is sensitive to inaccurate lead times, supplier delays, and production quality issues that may cause yield shortfalls .

# Agenda Course 4

- Production Scheduling
  - The role and benefits of scheduling
  - Simple sequencing rules
  - Sequencing theory for 1 machine
  - Sequencing for multiple machines
  - Line balancing
- Project Management
  - What is PM?
  - Major steps
  - Project scheduling : Gantt, CPM and PERT
  - Project costing

# Learning Objectives



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**Explain** the relationship between scheduling, capacity planning, MPS, ...

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**List** the different criteria out of which a schedule can be done

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**Describe** the priority sequencing rules

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**Apply** the assignment method for loading jobs

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**Use** Johnson's rule

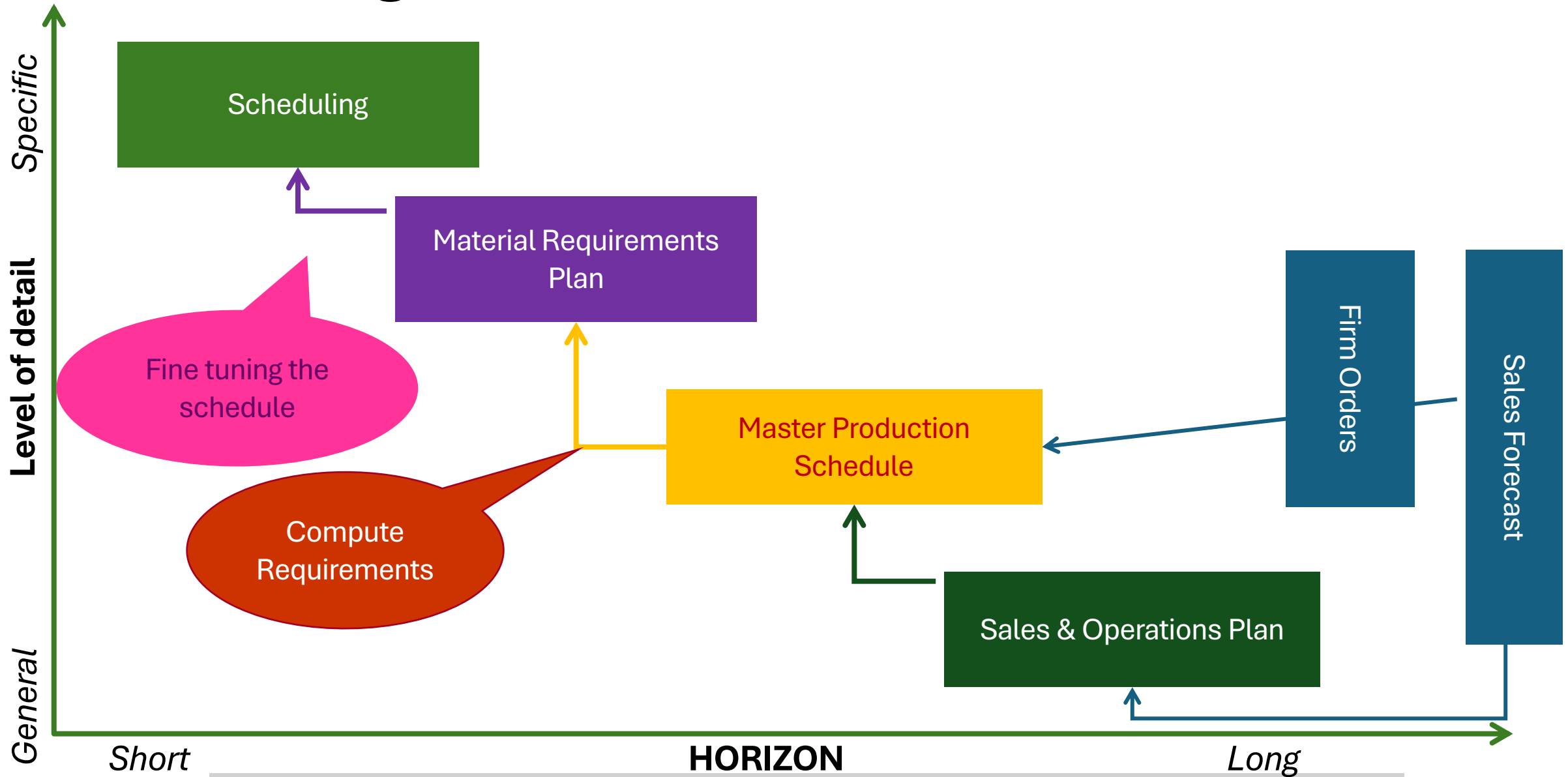
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**Define** finite capacity scheduling

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# Scheduling



# Production Scheduling

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Scheduling translates capacity decision, aggregate plans and MPS into job sequence and specific assignments of man, materials and machine

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Schedules deal with daily and even hourly assignments over a short period (1 or 2 weeks)

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Schedules need to be refreshed frequently to consider the latest information

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Efficient scheduling drives cost down and helps meet due date in an optimal way. It contributes to customer satisfaction.

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Scheduling can be done forward (hospitals, ...) or backward (manufacturing)

# What do we schedule?

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Departure time table, maintenance timetable

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Classrooms allocation, teacher schedule

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Production of goods, workers assignment

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Meal preparation

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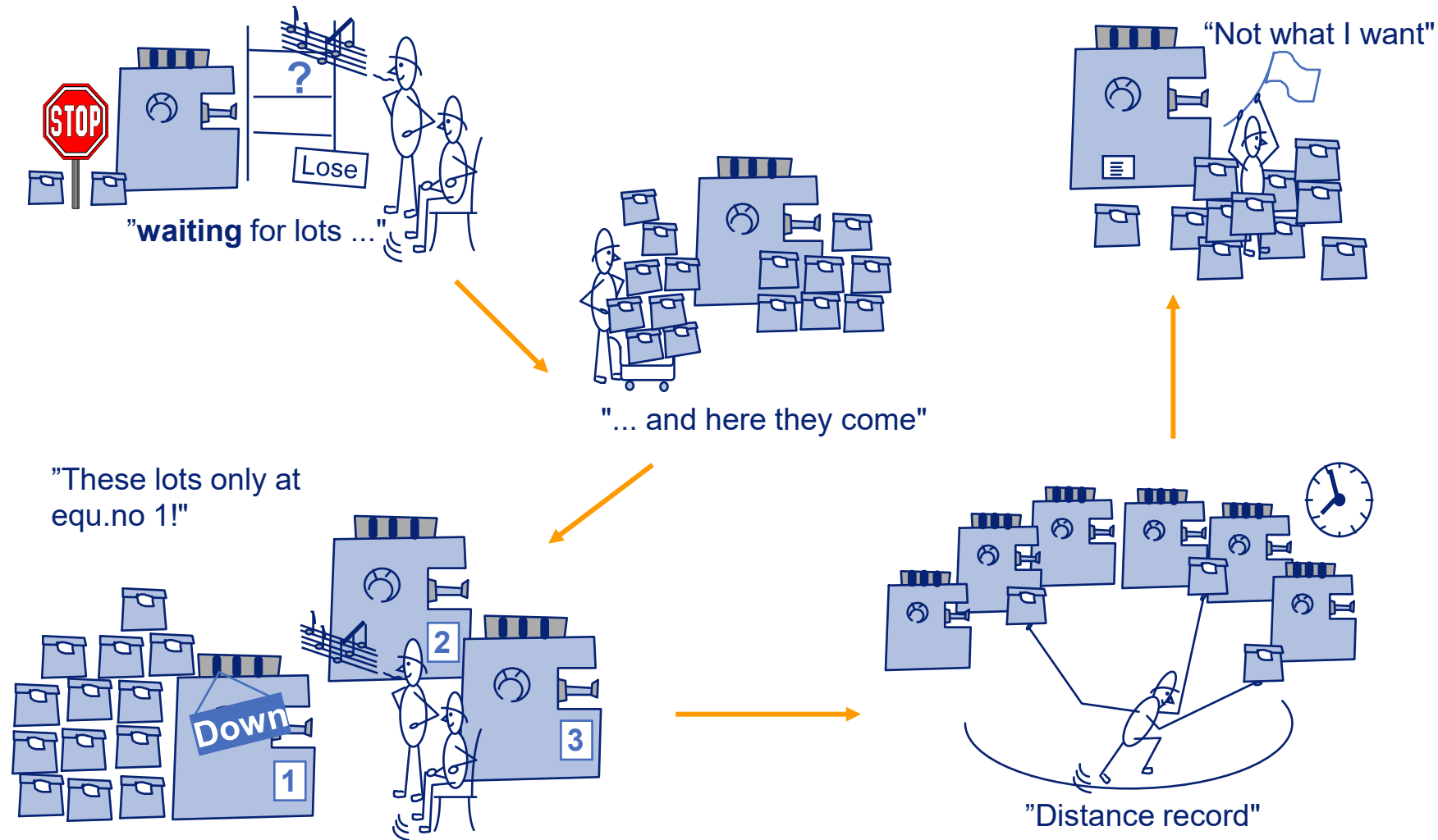
Projects

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Delivery routes

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# Without Production Scheduling





# With Production Scheduling



**Operator**

“I know which lots to retrieve next”



**Line Assistant**

“I know what the next setup will be”

**Line Planner**

“I can provide updated delivery forecasts”

**Manager**

“I am better prepared for potential issues in the line”

**Technician**

“I know when the next setup change will be”



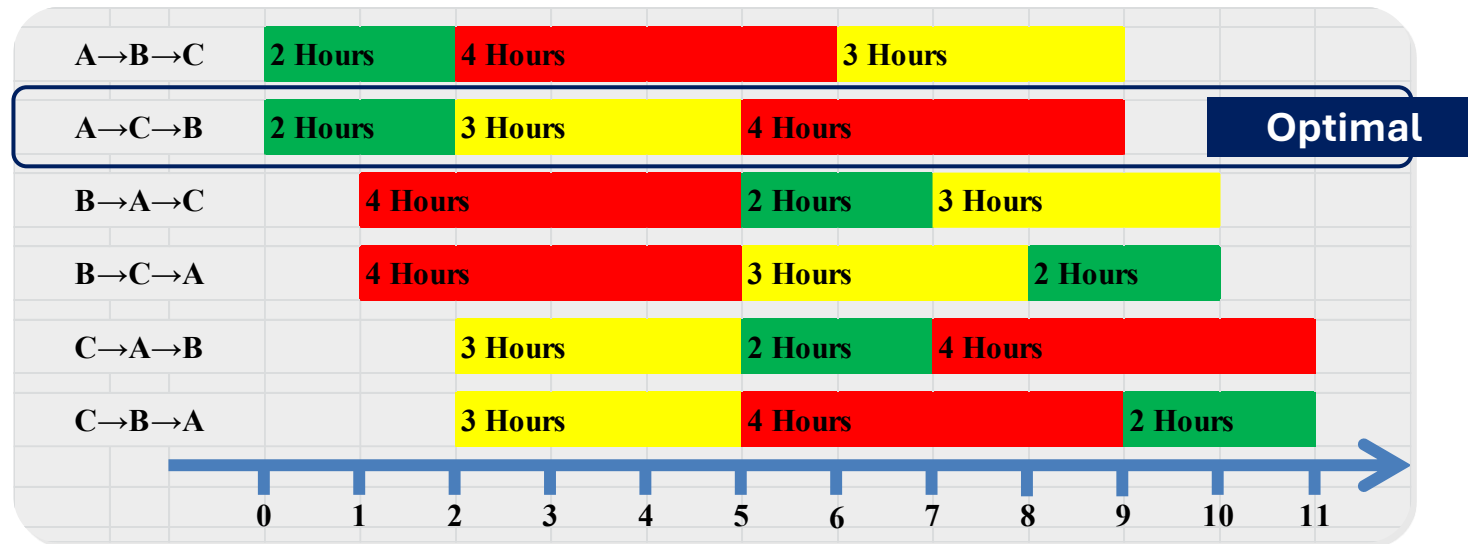
**Equipment Engineer**

“I know which equipment will be the bottleneck”

# Production Scheduling

**Problem:** Find the schedule for the 3 jobs on a single machine such that all the jobs can be completed as early as possible.

| Job | Cycle Time | Release Time | Due Time |
|-----|------------|--------------|----------|
| A   | 2 Hours    | 0            | 5        |
| B   | 4 Hours    | 1            | 9        |
| C   | 3 Hours    | 2            | 7        |



# Production Scheduling



```
tuple Job {
    string name;
    int p;
    int r;
    int d;
};

{Job} jobs = {<"A",2,0,5> , <"B",4,1,9>, <"C",3,2,7>};

int BigM = 1000000;

dvar int+ S[j in jobs];
dvar boolean Z[j1 in jobs][j2 in jobs];

minimize    sum(j in jobs) S[j];

subject to {

    forall(j in jobs){
        C1: S[j] + j.p <= j.d;
        C2: j.r <= S[j];
    };

    forall(j1 in jobs, j2 in jobs: j1 != j2){
        C3: S[j1] + j1.p <= S[j2] + (1 - Z[j1][j2]) * BigM;
        C4: S[j2] + j2.p <= S[j1] + Z[j1][j2] * BigM;
    };
};
```

Job A starts at 0 and ends at 2.  
Job B starts at 5 and ends at 9.  
Job C starts at 2 and ends at 5.

Define each Job with  
given characteristics

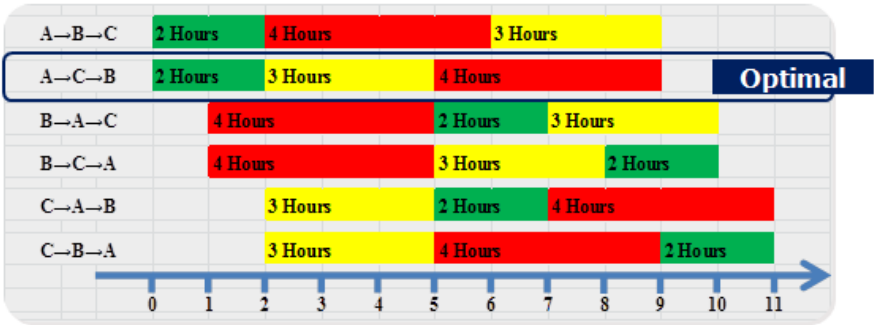
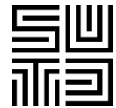
Decision  
variables

Objective is to  
minimize all starting  
times

Time constraints on  
each Job

Relative sequence  
of every two Jobs

# Production Scheduling



Job A starts at 0 and ends at 2.  
Job B starts at 5 and ends at 9.  
Job C starts at 2 and ends at 5.

Sequential



Simultaneous

High Scalability

# Production Scheduling: some definitions

We will focus on job shop scheduling where jobs to fulfill demand must be processed through  $n$  machines and more precisely on flow shops where each job goes through the same process. Job shop problems are extremely complex.

- Forward scheduling: job starts as soon as requirements are clear => what is the earliest end date?
- Backward scheduling: the due date is known => when should the job start to meet due date?
- Loading assigns the job to a work station
- Finite loading considers capacity constraints

Scheduling criteria:

- Minimize completion time: measured by average completion time
- Maximize utilization: measured by the % of time the facility is utilized
- Minimize WIP: measured by looking at the average number of jobs in the system
- Minimize customer waiting time: measured by the average number of late period
- Minimize the number of set-ups

# Production Scheduling: criteria

Scheduling criteria:

- Minimize completion time: measured by average completion time
- Maximize utilization: measured by the % of time the facility is utilized
- Minimize WIP: measured by looking at the average number of jobs in the system
- Minimize customer waiting time: measured by the average number of late period
- Minimize the number of set-ups

# Simple sequencing rules

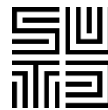
We focus on a **single machine** where **several jobs** need to be processed.

There are 4 rules used in practice:

- First Come First Served: **FCFS**
- Shortest Processing Time: **SPT**: jobs are sequenced in increasing order of their processing time
- Earliest Due Date: **EDD**: jobs are sequenced in increasing order of their due date
- Critical Ratio: **CR**: you form the ratio of the processing time over the remaining time till due date; the job with the largest ratio is scheduled next

We can evaluate their performance along:

- Mean Flow time: total flow time over number of jobs
- Average tardiness: Total late days over number of jobs
- Utilization: total processing time over total flow time



# Simple sequencing rules: example

Jobs arrived according to their letter.

| Job | Processing Time | Due Days |
|-----|-----------------|----------|
| A   | 6               | 8        |
| B   | 2               | 6        |
| C   | 8               | 18       |
| D   | 3               | 15       |
| E   | 9               | 23       |



# Simple sequencing rules: FCFS

Jobs arrived according to their letter.

| Job | Processing Time | Flow Time | Due Days | Lateness |
|-----|-----------------|-----------|----------|----------|
| A   | 6               | 6         | 8        | 0        |
| B   | 2               | 8         | 6        | 2        |
| C   | 8               | 16        | 18       | 0        |
| D   | 3               | 19        | 15       | 4        |
| E   | 9               | 28        | 23       | 5        |

- Mean Flow time:  $77/5=15,4$  days
- Average tardiness:  $11/5=2,2$
- Utilization:  $28/77=36,4\%$

# Simple sequencing rules: SPT

Jobs arrived according to their letter. SPT: Shortest Processing Time

| Job | Processing Time | Flow Time | Due Days | Lateness |
|-----|-----------------|-----------|----------|----------|
| B   | 2               | 2         | 6        | 0        |
| D   | 3               | 5         | 15       | 0        |
| A   | 6               | 11        | 8        | 3        |
| C   | 8               | 19        | 18       | 1        |
| E   | 9               | 28        | 23       | 5        |

- Mean Flow time: 13 days
- Average tardiness: 1,8 days
- Utilization: 43,1%

# Simple sequencing rules: EDD

Jobs arrived according to their letter.

| Job | Processing Time | Flow Time | Due Days | Lateness |
|-----|-----------------|-----------|----------|----------|
| B   | 2               | 2         | 6        | 0        |
| A   | 6               | 8         | 8        | 0        |
| D   | 3               | 11        | 15       | 0        |
| C   | 8               | 19        | 18       | 1        |
| E   | 9               | 28        | 23       | 5        |

- Mean Flow time: 13,6 days
- Average tardiness: 1,2 days
- Utilization: 41,2%

# Simple sequencing rules: CR

Jobs arrived according to their letter.

| t=0 Job | Processing Time | CR          | Due Days |
|---------|-----------------|-------------|----------|
| A       | 6               | $8/6=1,3$   | 8        |
| B       | 2               | $6/2=3$     | 6        |
| C       | 8               | $18/8=2,25$ | 18       |
| D       | 3               | $15/3=5$    | 15       |
| E       | 9               | $23/9=2,5$  | 23       |

=> Job A is 1st

| t=6 Job | Processing Time | CR         | Due Days |
|---------|-----------------|------------|----------|
|         |                 |            |          |
| B       | 2               | 0*         | 6        |
| C       | 8               | $12/8=1,5$ | 18       |
| D       | 3               | $9/3=3$    | 15       |
| E       | 9               | $17/9=1,9$ | 23       |

=> Job B is next

\* CR is 0 as we start at time 6

# Simple sequencing rules: CR

Jobs arrived according to their letter.

| t=8 Job | Processing Time | CR          | Due Days |
|---------|-----------------|-------------|----------|
| C       | 8               | $10/8=1,25$ | 18       |
| D       | 3               | $7/3=2,3$   | 15       |
| E       | 9               | $15/9=1,6$  | 23       |

=> Job C is next

| t=16 Job | Processing Time | CR     | Due Days |
|----------|-----------------|--------|----------|
| D        | 3               | $-1/3$ | 15       |
| E        | 9               | $7/9$  | 23       |

=> Job D is next then E

⇒ In our special case CR method is equivalent to FCFS

⇒ When  $CR < 1$  it is behind schedule and needs to be scheduled as a priority

# Sequencing theory for a single machine

n jobs to be scheduled by a single machine.

For each job i :

$t_i$  = Processing time

$d_i$  = Due date

$W_i$  = Waiting time

$F_i$  = Flowtime =  $W_i + t_i$

$L_i$  = Lateness =  $F_i - d_i$

$T_i$  = Tardiness =  $\max[L_i, 0]$

$E_i$  = Earliness =  $\max[-L_i, 0]$

Maximum Tardiness:

$T_{\max} = \max\{T_1, T_2, \dots, T_n\}$

Mean Flow time:

$$F' = \frac{1}{n} \sum_{i=1}^n F_i$$

Number of permutations:  $n!$

# Sequencing theory: theorem 1

The scheduling rule which minimizes mean flow time is the Shortest Processing Time. It also minimizes min waiting time and mean lateness.

Proof: Let  $[1], [2], \dots, [n]$  be a permutation;  
Flow time of job at position  $k$  is:

$$F_{[k]} = \sum_{i=1}^k t_{[i]}$$

Mean Flow time:

$$\begin{aligned} F' &= \frac{1}{n} \sum_{k=1}^n F_{[k]} = \frac{1}{n} \sum_{k=1}^n \sum_{i=1}^k t_{[i]} \\ &= \frac{1}{n} (nt_{[1]} + (n-1)t_{[2]} + \dots + t_{[n]}) \end{aligned}$$

Which is clearly minimized by setting:

$$t_{[1]} \leq t_{[2]} \leq \dots \leq t_{[n]}$$

# Sequencing theory: Moore algorithm

In business there are cases where the penalty is the same whatever the delay.

The algorithm **minimizes the number of tardy jobs.**

1. Sequence jobs in EDD order.
2. Find first tardy job, say  $[i]$ . If none, go to 4.
3. Consider jobs  $[1], [2], \dots, [i]$ , and reject job with largest processing time. Return to step 2.
4. Optimal sequence = current sequence with rejected jobs appended.



# Sequencing theory: Moore algorithm: example



| Job             | 1  | 2 | 3 | 4  | 5  | 6  |
|-----------------|----|---|---|----|----|----|
| Due date        | 15 | 6 | 9 | 23 | 20 | 30 |
| Processing time | 10 | 3 | 4 | 8  | 10 | 6  |

STEP 1/2: order by EDD, job 1 is tardy and has the longest processing time among (2,3,1) => reject job 1

| Job       | 2 | 3 | 1  | 5  | 4  | 6  |
|-----------|---|---|----|----|----|----|
| Flow Time | 3 | 7 | 17 | 27 | 35 | 41 |
| Due Date  | 6 | 9 | 15 | 20 | 23 | 30 |
| Lateness  | 0 | 0 | 2  | 7  | 12 | 11 |

STEP 3/2: job 4 is tardy, job 5 has the longest processing time => reject job 5

| Job       | 2 | 3 | 5  | 4  | 6  |  |
|-----------|---|---|----|----|----|--|
| Flow Time | 3 | 7 | 17 | 25 | 31 |  |
| Due Date  | 6 | 9 | 20 | 23 | 30 |  |
| Lateness  | 0 | 0 | 0  | 2  | 1  |  |

# Sequencing theory: Moore algorithm: example



| Job             | 1  | 2 | 3 | 4  | 5  | 6  |
|-----------------|----|---|---|----|----|----|
| Due date        | 15 | 6 | 9 | 23 | 20 | 30 |
| Processing time | 10 | 3 | 4 | 8  | 10 | 6  |

STEP 3/2:

| Job       | 2 | 3 | 4  | 6  |  |  |
|-----------|---|---|----|----|--|--|
| Flow Time | 3 | 7 | 15 | 21 |  |  |
| Due Date  | 6 | 9 | 20 | 23 |  |  |
| Lateness  | 0 | 0 | 0  | 0  |  |  |

STEP 3/2/4:

| Job       | 2 | 3 | 4  | 6  | 1  | 5  |
|-----------|---|---|----|----|----|----|
| Flow Time | 3 | 7 | 15 | 21 | 31 | 39 |
| Due Date  | 6 | 9 | 20 | 23 | 15 | 23 |
| Lateness  | 0 | 0 | 0  | 0  | 16 | 16 |

# Sequencing theory: Lawler algorithm

The algorithm **minimizes the tardiness**.

- Protects jobs with **tight deadlines**
- Delays “less urgent” jobs safely

The algorithm schedules first the job to be completed last and so on. At each stage it determines the set of jobs,  $V$ , not required to precede any remaining job. As constraints might evolve the set  $V$  might change as we go along.

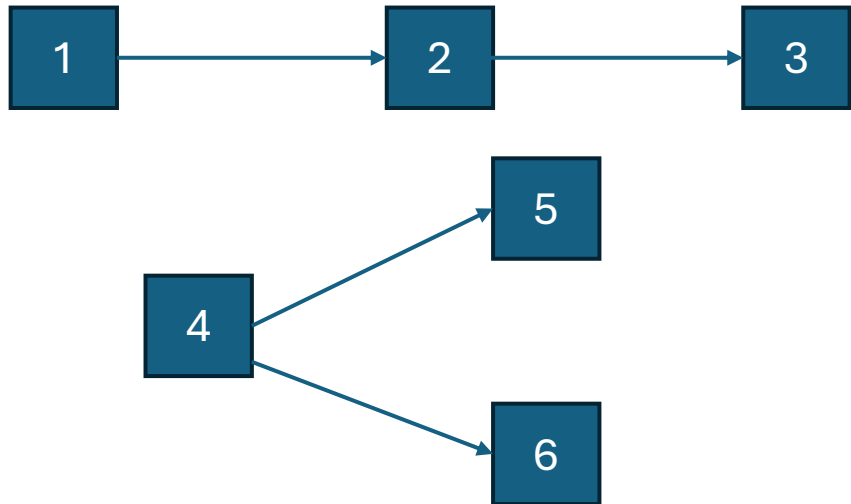
- 1- Draw the precedence constraint graph
  - 2- Set  $V$  the jobs not required to precede others
  - 3- Choose job  $k$  such as:
    - $g_k(\tau) = \min (g_i(\tau))$  where  $i \in V$
    - and  $\tau$  is the processing time of the current sequence
    - $\tau = \sum t_i$  from 1 to  $n$
    - $k$  is now scheduled last
- Continue till all jobs are scheduled

# Sequencing theory: Lawler algorithm: example



At the automotive painting and repair shop:

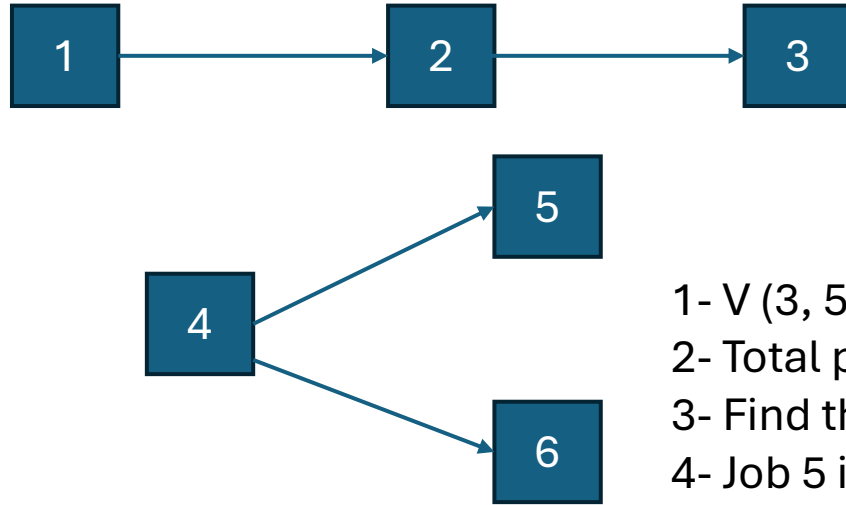
- 6 cars
- Cars 1, 2, 3 to finished at the date they were promised
- Cars 4,5,6 where 4 is to be completed 1st



| Job   | 1 | 2 | 3 | 4 | 5  | 6 |
|-------|---|---|---|---|----|---|
| $t_i$ | 2 | 3 | 4 | 3 | 2  | 1 |
| $d_i$ | 3 | 6 | 9 | 7 | 11 | 7 |

$t_i$ : processing time for job  $i$   
 $d_i$ : due date for job  $i$

# Sequencing theory: Lawler algorithm: example



| Job   | 1 | 2 | 3 | 4 | 5  | 6 |
|-------|---|---|---|---|----|---|
| $t_i$ | 2 | 3 | 4 | 3 | 2  | 1 |
| $d_i$ | 3 | 6 | 9 | 7 | 11 | 7 |

- 1-  $V(3, 5, 6)$ : jobs not required to precede any other
- 2- Total processing time:  $\tau = 15$
- 3- Find the job  $k$  out of  $V$  with the smallest tardiness ( $g_k(\tau) = \min(g_i(\tau))$ ): this is **job 5**
- 4- Job 5 is scheduled last
- 5- Reiterate:
  - Total processing time : 13
  - min tardiness (job 3, 6): **job 3**
- 6- The new set is  $V(2, 6)$ :
  - Total remaining processing time : 9
  - min tardiness (job 2, 6): **job 6**
- 7- The new set is  $V(4, 2)$ 
  - Total remaining processing time: 8
  - min tardiness (job 4, 2): **job 4**
- 8- Next is 2 then 1 with a simple precedence rule
- 9- The sequence is thus 1 2 4 6 3 5

| Job | Due Date | Tardiness |
|-----|----------|-----------|
| 1   | 3        | 0         |
| 2   | 6        | 0         |
| 4   | 7        | 1         |
| 6   | 7        | 2         |
| 3   | 9        | 4         |
| 5   | 11       | 4         |

# Sequencing theory for multiple machines

We increase the complexity of  $n$  jobs on  $m$  machines.

There are  $(n!)^m$  permutations.

For a case as simple as 5 jobs on 5 machines we have 25 billions possible schedule.

An attractive way to represent a schedule is to use Gantt charts.

We will focus on solutions for scheduling  **$n$  jobs on 2 machines**, with the objective to **minimize the total flow time**.

Theorem: the optimal solution for scheduling  $n$  jobs on 2 machines is always a permutation schedule. This is to say the jobs follow the same order on both machines.

=> Johnson algorithm for solving this 2 machines problem.

# Johnson algorithm for n jobs on 2 machines

2 machines A and B.

All jobs start on machine A. There are n jobs.

$A_i$  = processing time of job  $i$  on  $A$

$B_i$  = processing time of job  $i$  on  $B$

*Basic Rule:* Job  $i$  precedes  $j$  if  $\min(A_i, B_j) < \min(A_j, B_i)$

- ① List values of  $A_i$  and  $B_i$  in 2 columns
- ② Find smallest remaining value in the 2 columns  
If it belongs to Col A, Schedule job next  
If it belongs to Col B, Schedule job last
- ③ Cross off scheduled job
- ④ Stop when done

# Johnson algorithm: example

The min job time is 1 for job 2 on machine A.  
 Thus job 2 is scheduled 1st.  
 The next min job time is 2 for job 1 on machine B. So job 1 is scheduled last.  
 Next is job 4 on machine A. Thus job 4 is scheduled 3rd.  
 Then it is job 5 on machine B. So job 5 is scheduled before last.  
 This gives 2, 4, 3, 5,1.  
 Below Gantt chart shows no idle time on A => optimal schedule.

| Job | Machine A | Machine B |
|-----|-----------|-----------|
| 1   | 5         | 2         |
| 2   | 1         | 6         |
| 3   | 9         | 7         |
| 4   | 3         | 8         |
| 5   | 10        | 4         |





# Akers procedure for 2 jobs on m machines



Each job must be processed in a particular order on all machines (route) but the 2 jobs need not have the same route.

- 1- Draw a cartesian coordinate system with job 1 on axis X and job 2 on axis Y and the coordinate the processing time
- 2- Blockout rectangular areas where both jobs would be processed at the same time
- 3- Draw a path from origin to the top right corner:
  - horizontal lines : job 1 is processed
  - vertical lines: job 2 is processed
  - 45deg lines: both jobs are processed at the same time on different machines

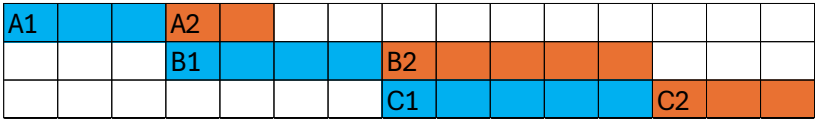
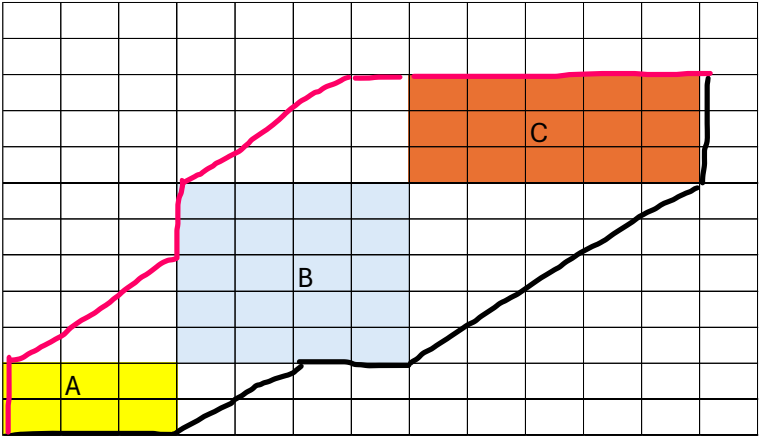
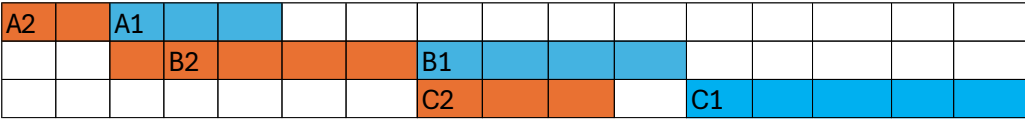
The path with the minimum distance is the optimal solution

# Akers procedure: example

2 jobs with same routes: A->B->C.  
 Tool time is calculated by the total time on horizontal axis (12) plus the total vertical movement (4 in blue and 3 in red)

| Job 1     |      | Job 2     |      |
|-----------|------|-----------|------|
| Operation | Time | Operation | Time |
| A         | 3    | A         | 2    |
| B         | 4    | B         | 5    |
| C         | 5    | C         | 3    |

Red Path Total time = 12+4



Black Path Total time = 12+3

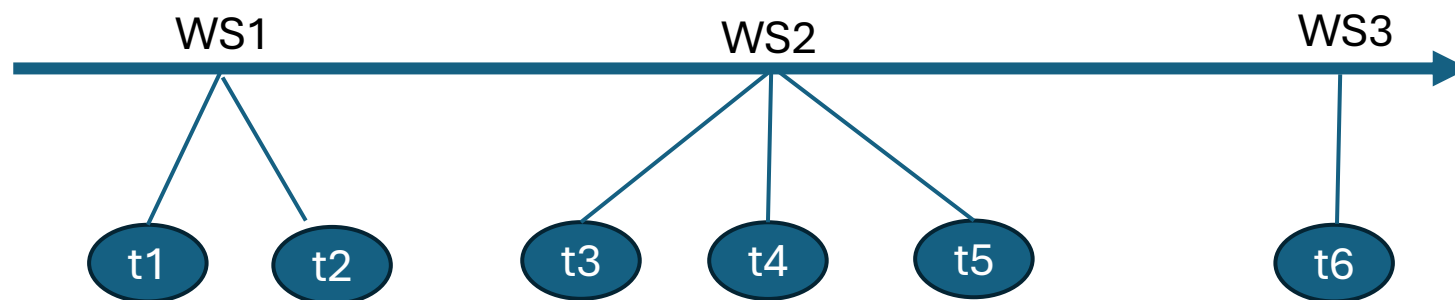
# Assembly Line Balancing

Classical Industrial Engineering problem.

We have a set of  $n$  distinct tasks to be performed on each item. The time to complete a task  $i$  is known and constant  $t_i$ .

The goal is to organize the tasks into groups performed on a workstation. The amount of time allocated to each workstation is determined in advance, based on the desired rate of production of the line. This is called cycle time  $C$ . We have precedence constraints and some tasks cannot be performed at the same workstation.

Schematically it looks like below:



# Assembly Line Balancing

There are several heuristics model to determine the appropriate balance.

Our objective is a cycle time  $C$ .

We have a total processing time :

$$T = \sum_{i=1}^n t_i$$

The number of WorkStations is the closest integer of  $T/C$ .

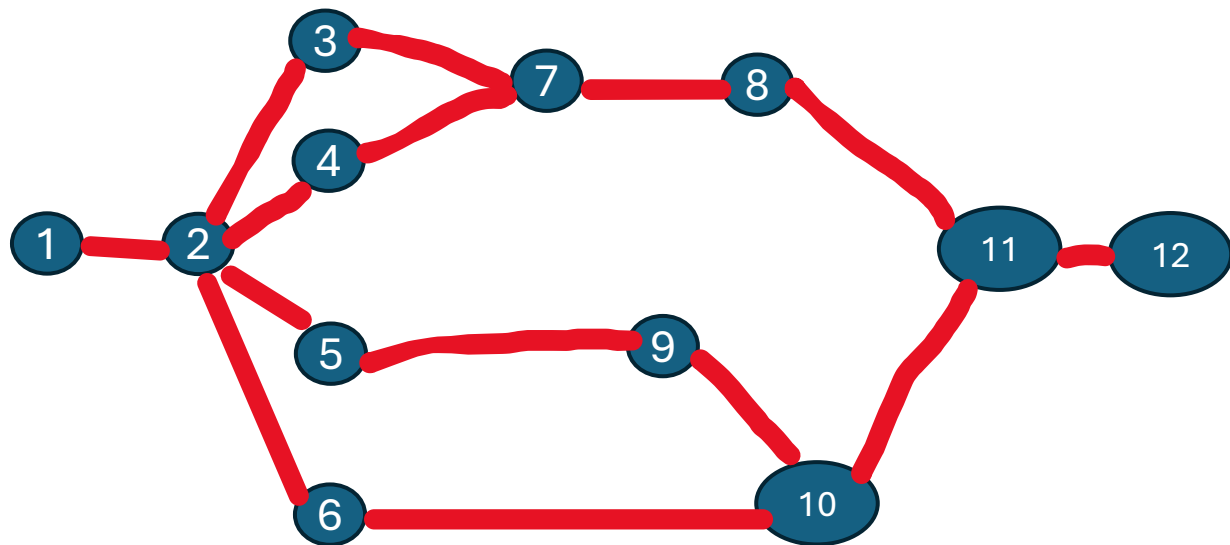
We will look at the method ‘ranked positional weight technique’. The method assigned a weight to each task based on the total time required by all succeeding tasks. Tasks are assigned sequentially to stations based on these weights.

- ① Determine the precedence network of tasks
- ② Positional weight of  $i = t_i + \sum_{j \in \text{succ}(i)} t_j$
- ③ Sort tasks in decreasing positional weight
- ④ Assume a “good” cycle-time  $C$
- ⑤ Assign jobs from sorted list without violating capacity and precedence
- ⑥ Try another  $C$ , i.e. go to 4.
- ⑦ Stop, when satisfactory cycle-time and balance is obtained.

# Assembly Line Balancing: example

An assembly line requires 12 tasks.

The precedence of each task is described in the table.



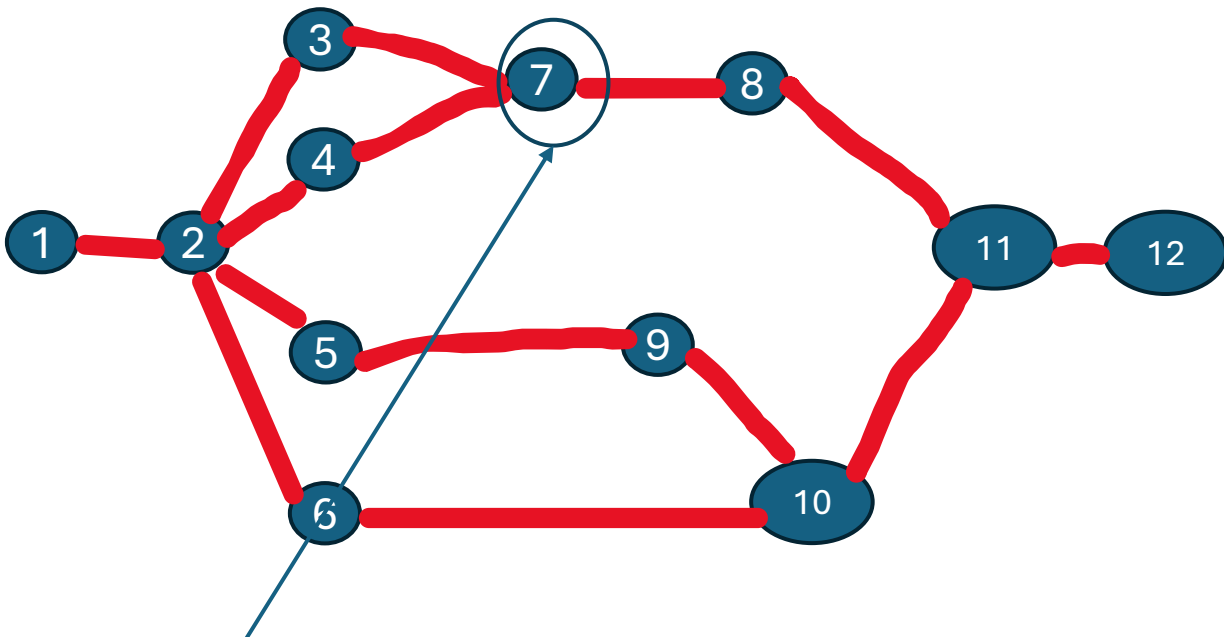
| Task | Immediate predecessors | Time |
|------|------------------------|------|
| 1    |                        | 12   |
| 2    | 1                      | 6    |
| 3    | 2                      | 6    |
| 4    | 2                      | 2    |
| 5    | 2                      | 2    |
| 6    | 2                      | 12   |
| 7    | 3,4                    | 7    |
| 8    | 7                      | 5    |
| 9    | 5                      | 1    |
| 10   | 9,6                    | 4    |
| 11   | 8,10                   | 6    |
| 12   | 11                     | 7    |

# Assembly Line Balancing: example

We want to produce one item every 15 min.

The number of WS is thus  $70/15 \sim 5$ .

The positional weight of task  $i$  is the time required to perform task  $i$  plus the time required to perform all tasks having a predecessor as  $i$ .



= time to perform 7 + time to perform 8 + time to perform 11 +  
time to perform 12

**=25**

| Task | Immediate predecessors | Time | P weight |
|------|------------------------|------|----------|
| 1    |                        | 12   | 70       |
| 2    | 1                      | 6    | 58       |
| 3    | 2                      | 6    | 31       |
| 4    | 2                      | 2    | 27       |
| 5    | 2                      | 2    | 20       |
| 6    | 2                      | 12   | 29       |
| 7    | 3,4                    | 7    | 25       |
| 8    | 7                      | 5    | 18       |
| 9    | 5                      | 1    | 18       |
| 10   | 9,6                    | 4    | 17       |
| 11   | 8,10                   | 6    | 13       |
| 12   | 11                     | 7    | 7        |

# Assembly Line Balancing: example

We then rank the tasks by decreasing P weight: 1 2 3 6 4  
7 5 8 9 10 11 12.

Tasks are now assigned sequentially without breaking the precedence rule.

Each WS has a 15min CT

WS 1 : task 1 idling 3 min (CT-processing time)

WS 2: tasks 2, 3, 4 idling 1min

WS 3: tasks 5, 6, 9 no idling

WS 4: tasks 7, 8 idling 3min

WS 5 : tasks 10, 11, idling 5 min

WS 6: task 12, idling 8min

We cannot fulfill the 5 WS target with C = 15min.

Thus we have to decide for another CT if we insist on 5 WS.

| Task | P weight | Time | Precedence | WS |
|------|----------|------|------------|----|
| 1    | 70       | 12   |            | 1  |
| 2    | 58       | 6    | 1          | 2  |
| 3    | 31       | 6    | 2          | 2  |
| 6    | 29       | 12   | 2          | 3  |
| 4    | 27       | 2    | 2          | 2  |
| 7    | 25       | 7    | 3,4        | 4  |
| 5    | 20       | 2    | 2          | 3  |
| 8    | 18       | 5    | 7          | 4  |
| 9    | 18       | 1    | 5          | 3  |
| 10   | 17       | 4    | 9,6        | 5  |
| 11   | 13       | 6    | 8,10       | 5  |
| 12   | 1        | 7    | 11         | 6  |

# Assembly Line Balancing: example

We decide on a CT of 16

WS 1 : task 1 idling 4 min (CT-processing time)

WS 2: tasks 2, 3, 4,5, no idling

WS 3: tasks 6,9, idling 3

WS 4: tasks 7,8,10, no idling

WS 5 : tasks 11,12, idling 3 min

More efficient balance:

- Idle time is reduced to 10
- 1 workstation less
- But a higher cycle time

If the factory runs 7 hours with a CT of 15min this is 28 units per assembly operation. If CT is 16min it reduces to 26,25.

=> Is it ok?

| Task | P weight | Time | Precedence | WS |
|------|----------|------|------------|----|
| 1    | 70       | 12   |            | 1  |
| 2    | 58       | 6    | 1          | 2  |
| 3    | 31       | 6    | 2          | 2  |
| 6    | 29       | 12   | 2          | 3  |
| 4    | 27       | 2    | 2          | 2  |
| 7    | 25       | 7    | 3,4        | 4  |
| 5    | 20       | 2    | 2          | 2  |
| 8    | 18       | 5    | 7          | 4  |
| 9    | 18       | 1    | 5          | 3  |
| 10   | 17       | 4    | 9,6        | 4  |
| 11   | 13       | 6    | 8,10       | 5  |
| 12   | 1        | 7    | 11         | 5  |





# Assembly Line Balancing: example

Another way to solve the plan: we go for 6 WS.

We find this is solvable with 13min CT

WS 1 : task 1 idling 1 min

WS 2: tasks 2, 3, idling 1 min

WS 3: tasks 6, idling 1 min

WS 4: tasks 4,5,7,9, idling 1 min

WS 5: tasks 8, 10, idling 4 min

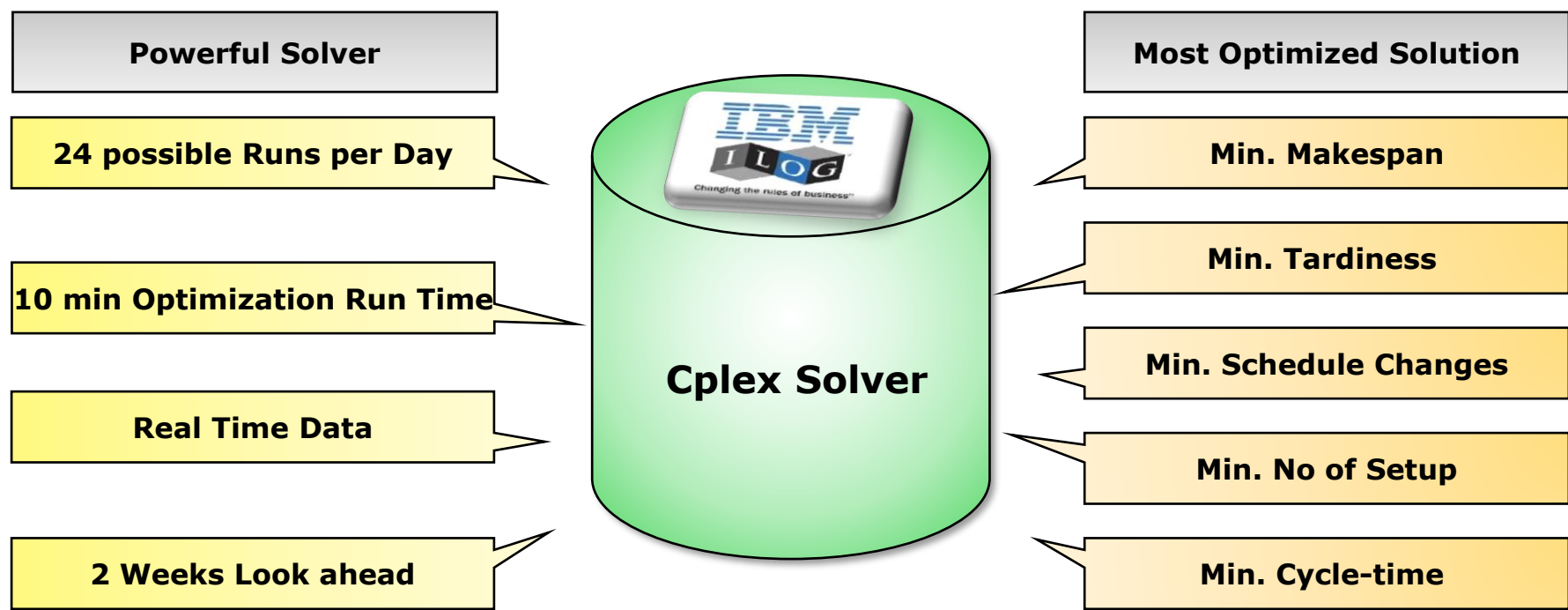
WS 6 : tasks 11,12, no idling

More WS better throughput in that case.

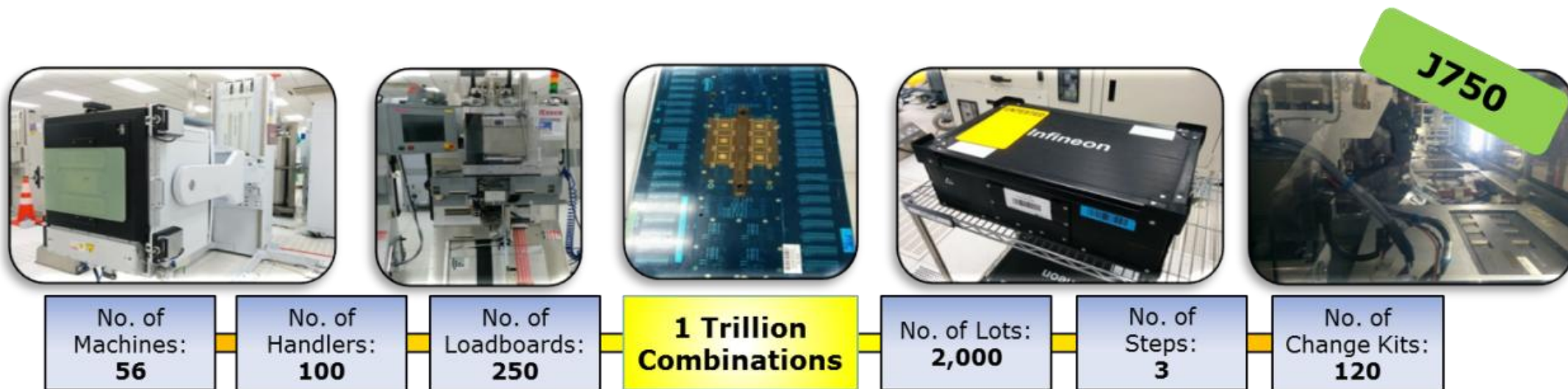
# Assembly Line Balancing and scheduling

There are other methods to consider but they would require complex and time consuming calculations.

More to come in our dedicated seminar.



# Real-life complexity: scheduling the test shopfloor at Infineon



# Wrap-up



Scheduling is the last ‘planning’ layer.

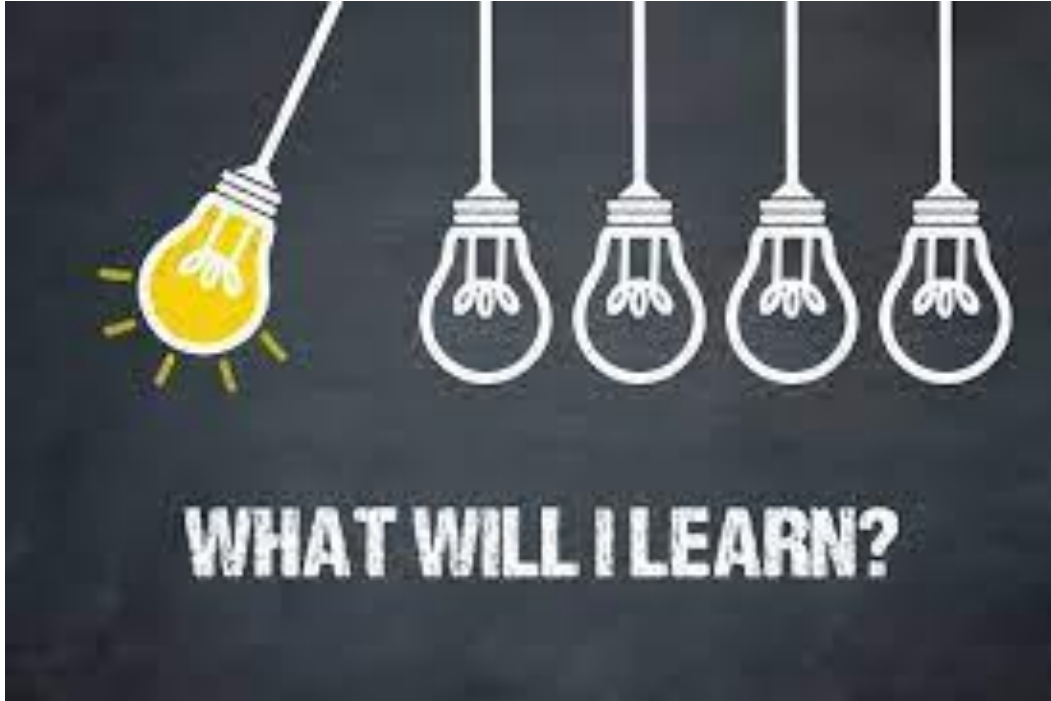
Its purpose is to handle the execution of the plan in the most effective and efficient way.

There is a wide variety of method, to be selected upon the objectives.

# Agenda Course 4

- Production Scheduling
  - The role and benefits of scheduling
  - Simple sequencing rules
  - Sequencing theory for 1 machine
  - Sequencing for multiple machines
  - Line balancing
- Project Management
  - What is PM?
  - Major steps
  - Project scheduling : Gantt, CPM and PERT
  - Project costing

# Learning Objectives



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**Use a Gantt diagram** for project planning

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---

**Use** AOA and AON networks for project representation

---

---

**Complete** forward and backward passes for a project

---

---

Understand the **Critical Path**

---

---

Evaluate the impact to the project due to **variability**

---

---

**Cost** a project

---

# What is project management

A project is a serie of connected tasks directed to a major outcome.

In current times projects can last a few months to several years. For large projects there will be a dedicated project organization to which resources are allocated.

There are 3 phases to project management:

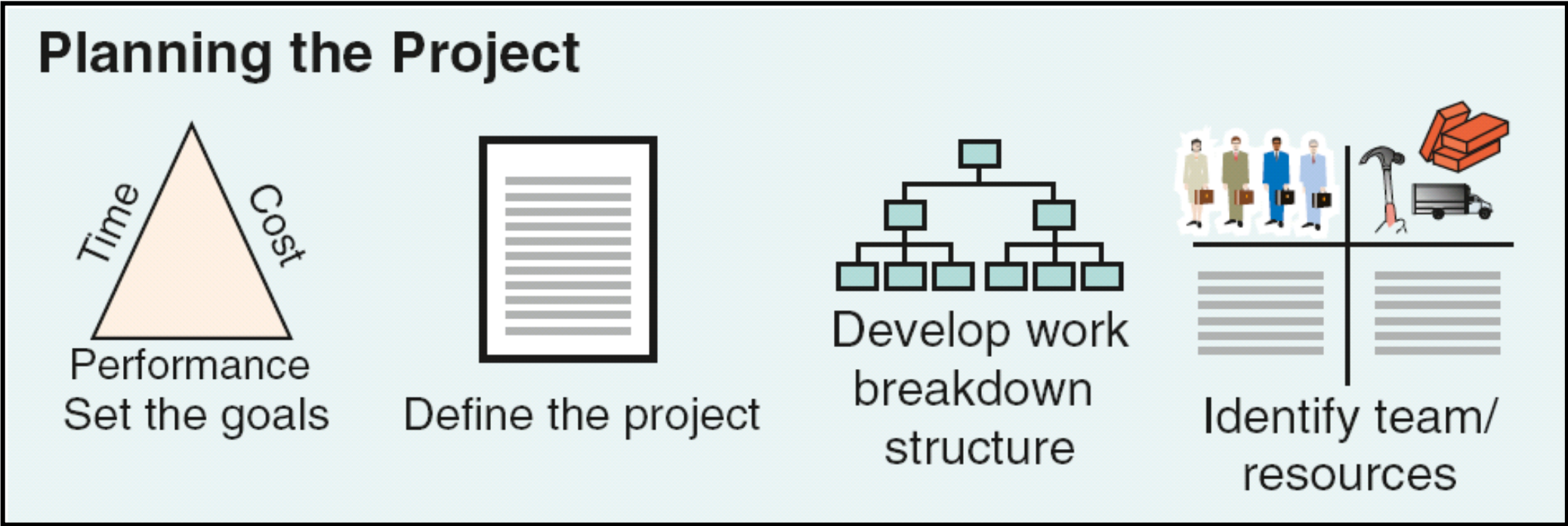
- Planning
- Scheduling
- Controlling

# Questions we have to answer

1. Which activities are covered under the project?
2. How to plan the project?
3. Which tools do I have for project planning?
4. What are the critical activities?
5. What is the probability to complete the project timely?
6. What are the risks related to the project?



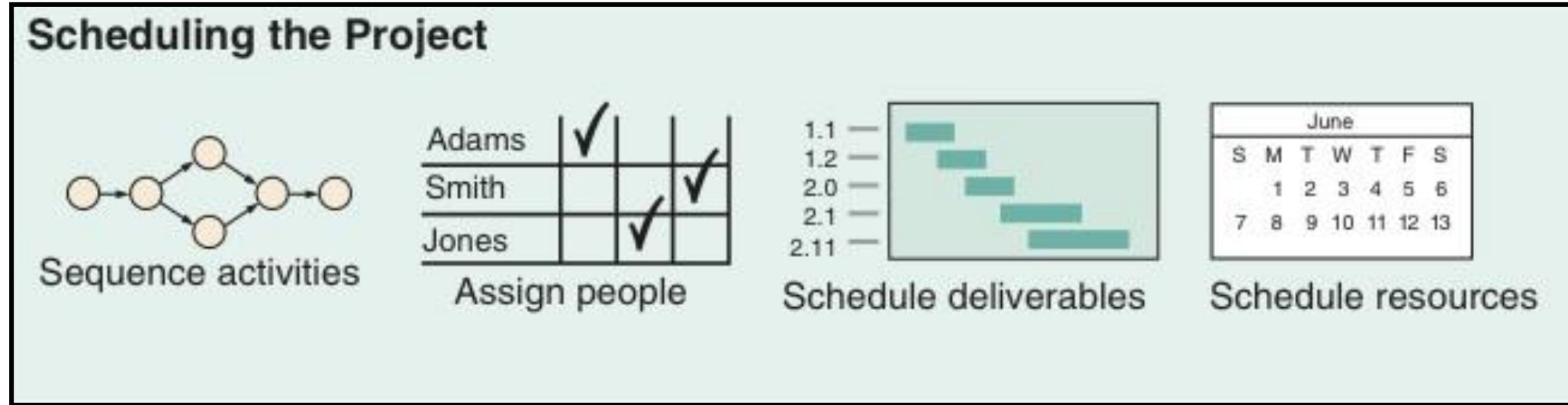
# Project phase 1: planning



Ch03, figure 3.1, Heizer, Render et Munson 2019

Define goals, clarify the project scope, allocate resources

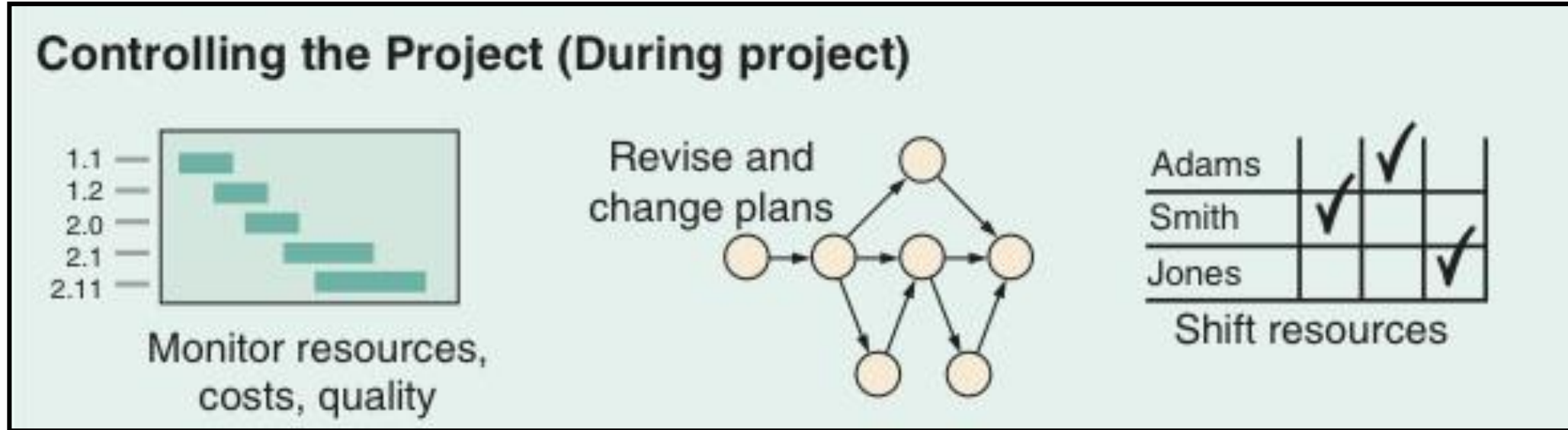
# Project phase 2: scheduling



Ch03, figure 3.1, Heizer, Render et Munson 2019

We link activities to people/supplier/budget  
We link activities together

# Project phase 6: controlling

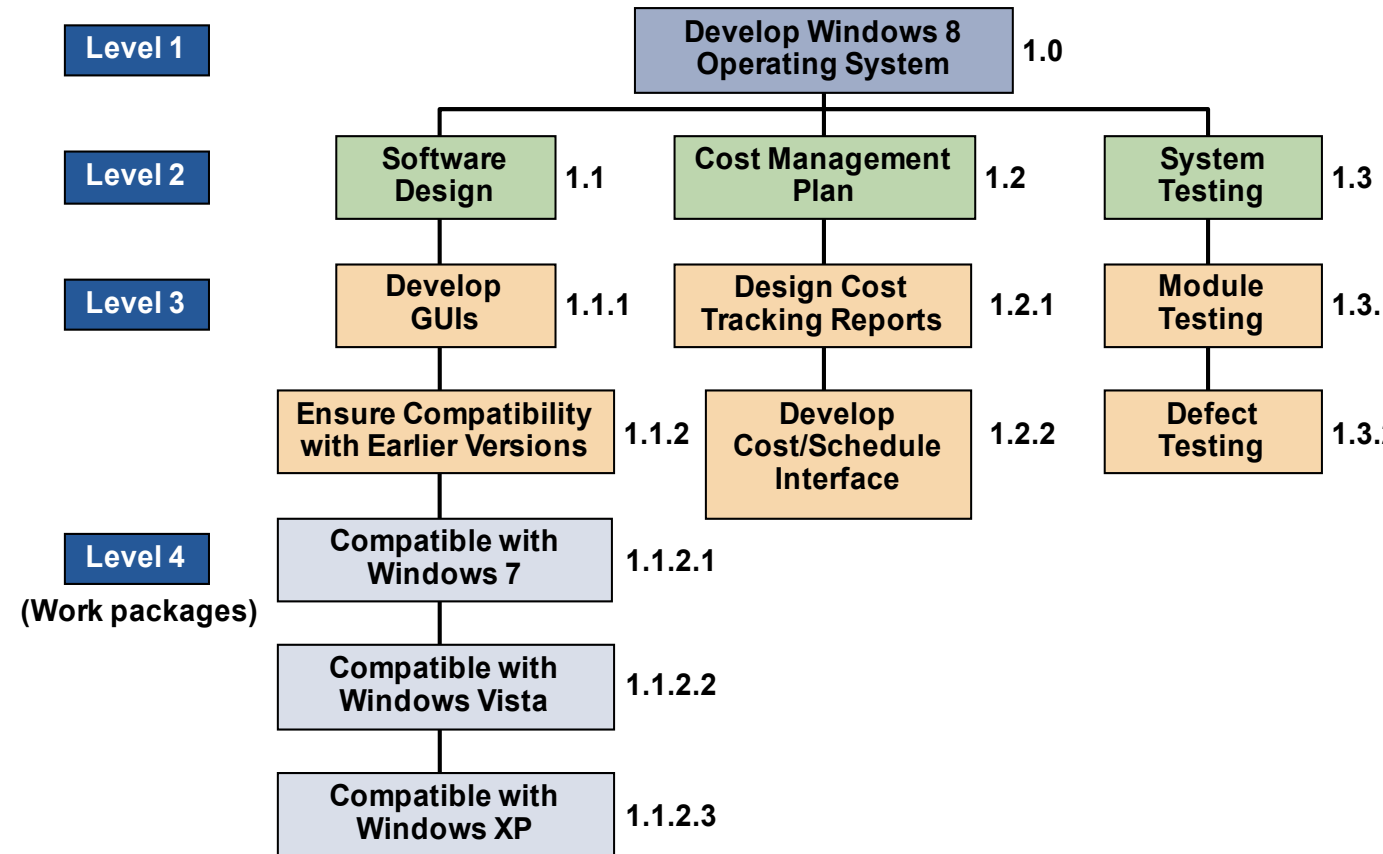


Ch03, figure 3.1, Heizer, Render et Munson 2019

We track progress (budget, timing, resources); we update plans; we reallocate resources; we mitigate risk

# Work Breakdown Structure (WBS)

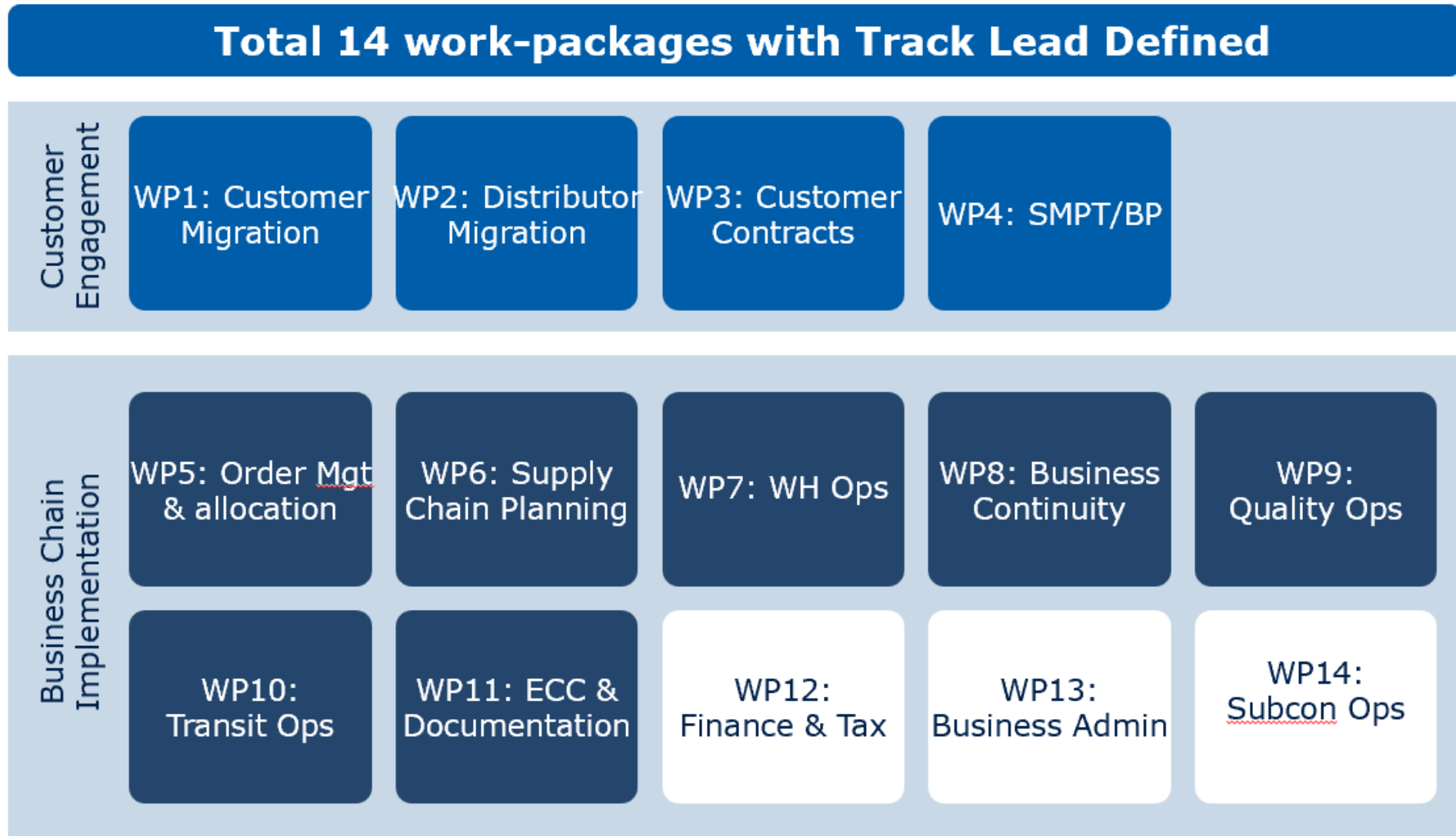
The project is broken down into manageable parts, the WBS.  
 Each part is further broken down in components.  
 Each component is further broken down into activities or work packages.  
 This decomposition exercise might be difficult but is necessary to manage the project efficiently.



Ch03, figure 3.3, Heizer, Render et Munson 2019

**An example for the development of Microsoft Windows 8**

# Work Breakdown Structure (WBS): example



# New approach to PM: Agile method

| Traditional                        | Agile                             |
|------------------------------------|-----------------------------------|
| Design up front                    | Continuous design                 |
| Fixed scope                        | Flexible                          |
| Deliverables                       | Features/requirements             |
| Freeze design as early as possible | Freeze design as late as possible |
| Low uncertainty                    | High uncertainty                  |
| Avoid change                       | Embrace change                    |
| Low customer interaction           | High customer interaction         |
| Conventional project teams         | Self-organized project teams      |

# Project scheduling

When preparing our schedule we need to ensure:

1. All activities are planned
2. Their sequence is documented
3. The duration of each activity is recorded
4. The total duration of the project is estimated

Our goals are:

1. Show how activities are linked to each other and to the overall project outcome
2. Identify priorities among activities
3. Be realistic in our time and cost evaluation
4. Manage scarce resources effectively and efficiently



# Poor Project scheduling

The Elbphilharmonie was supposed to open in 2010 - but was finished seven years later, even though construction workers sometimes worked around the clock. Initial construction costs were estimated at **77 million euros (\$85.6 million)**, but final costs exceeded 10 times that amount.



**OTHER EXAMPLES??**

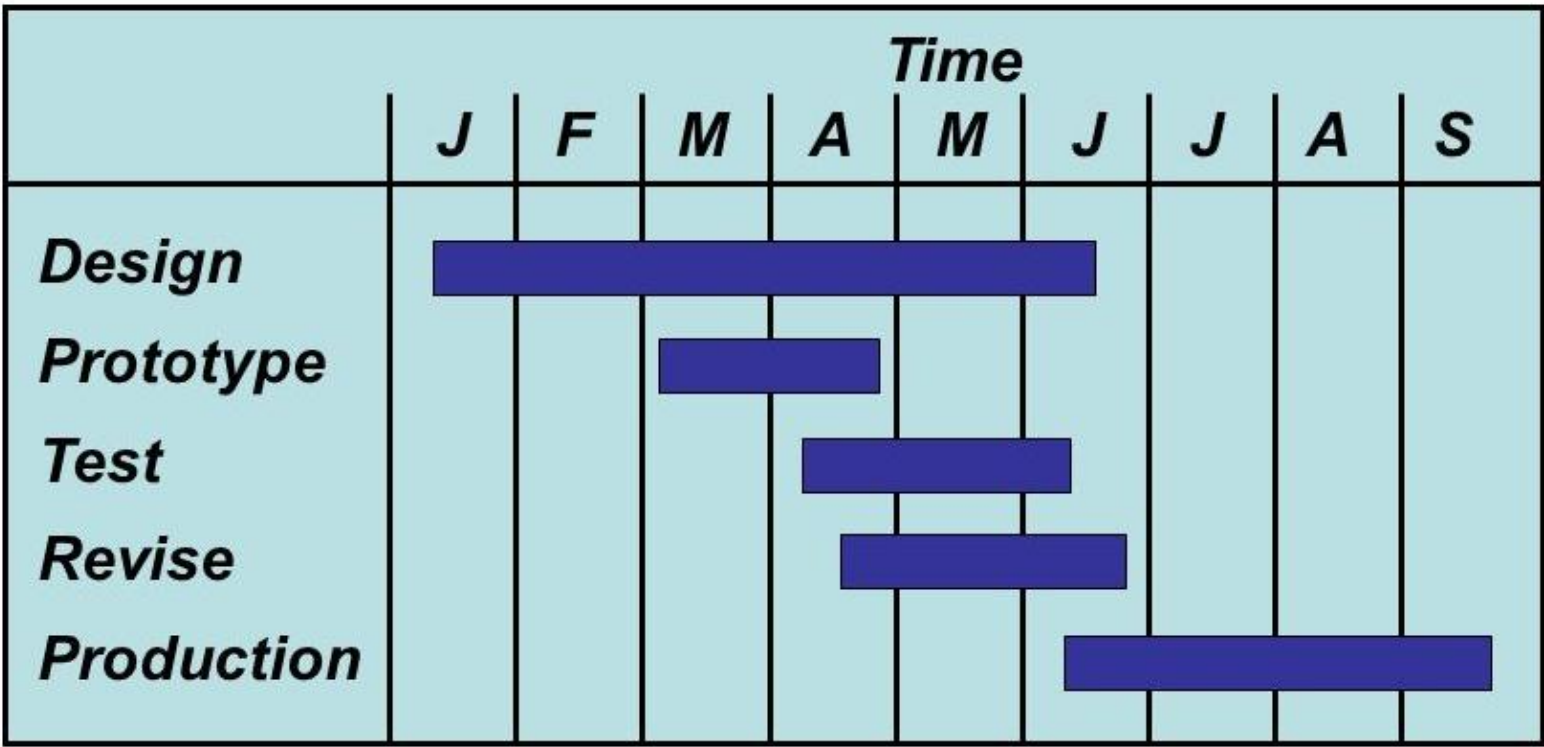


# Project scheduling methods

3 different methods are used:

1. Gantt charts: the most common and simplest to represent the different activities, their duration and order; it lacks the ability to link activities with one another
2. Network charts: the project is modelled as a network to represent precedence relationship and interdependencies:
  - Critical Path Method for deterministic project: **CPM**
  - Project Evaluation and Review Technique to account for some randomness in duration: **PERT**

# Project scheduling: Gantt



# Project scheduling: CPM and PERT

6 steps to follow:

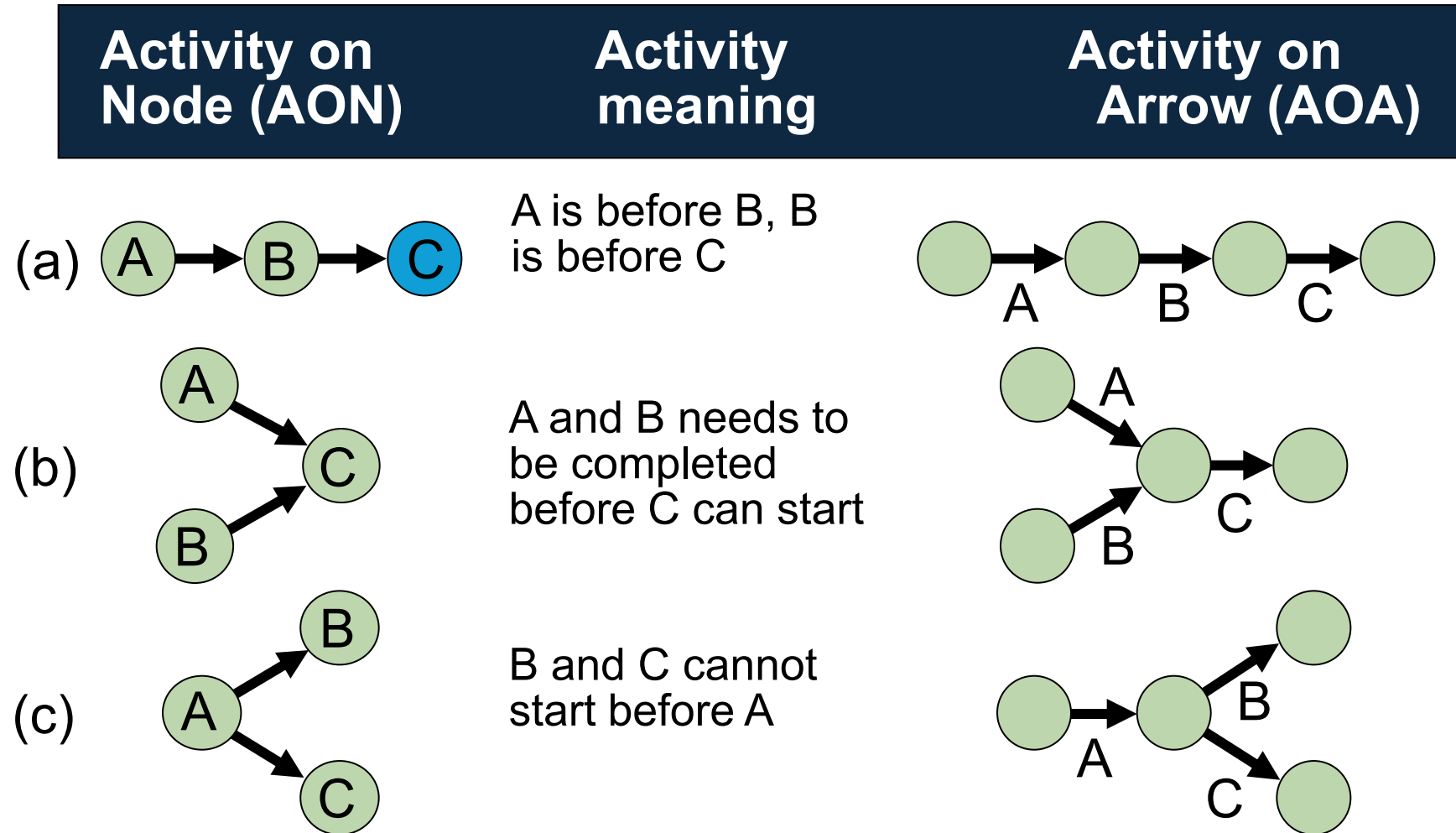
- Define the project and prepare the WBS
- Develop the relationship between activities
- Draw the network linking all activities
- Assign time to each activity
- Compute the longest time path through the network: this is our critical path
- Use the network to plan, schedule, monitor, amend and control the project

# Project scheduling: CPM and PERT

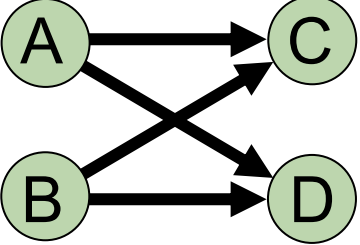
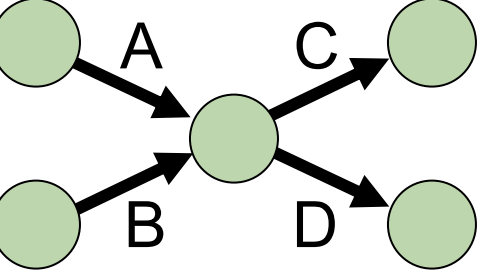
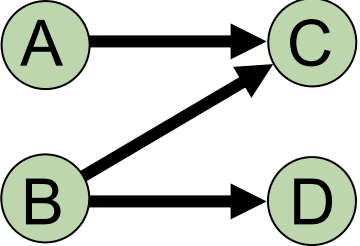
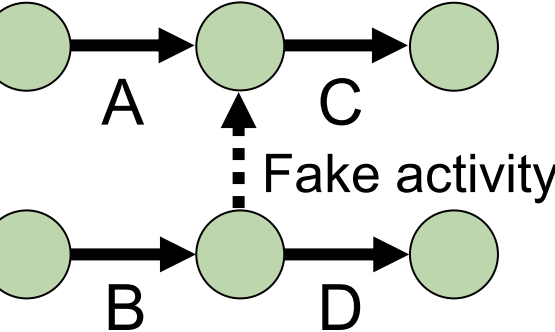
Key questions PERT and CPM are helping us with:

- When will the entire project be completed
- What are the critical activities which might delay the whole schedule
- What are the non critical activities
- What is the probability the project will be completed on time (for PERT only)
- Is the project on schedule and on budget or not
- Are there enough resources
- Trade-offs of expediting tasks to complete the project earlier

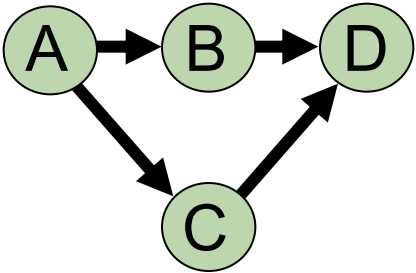
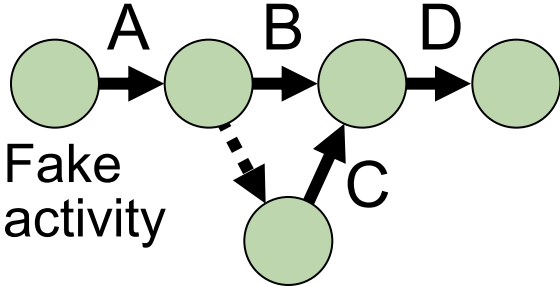
# Network representation: AON or AOA



# Network representation: AON or AOA

| Activity on Node (AON)                                                                 | Activity meaning                                                                                              | Activity on Arrow (AOA)                                                              |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| (d)   | C et D cannot start as long as A and B are not completed                                                      |   |
| (e)  | C cannot start till A and B are completed<br>D cannot start before B<br>A fictive activity is deployed in AOA |  |

# Network representation: AON or AOA

| Activity on Node (AON)                                                                       | Activity meaning                                                                                                                                | Activity on Arrow (AOA)                                                             |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p>(F)</p>  | <p>B and C cannot start till A is completed</p> <p>D cannot start till B and C are completed</p> <p>A fictive activity is introduced in AOA</p> |  |

# Example: Milwaukee Paper

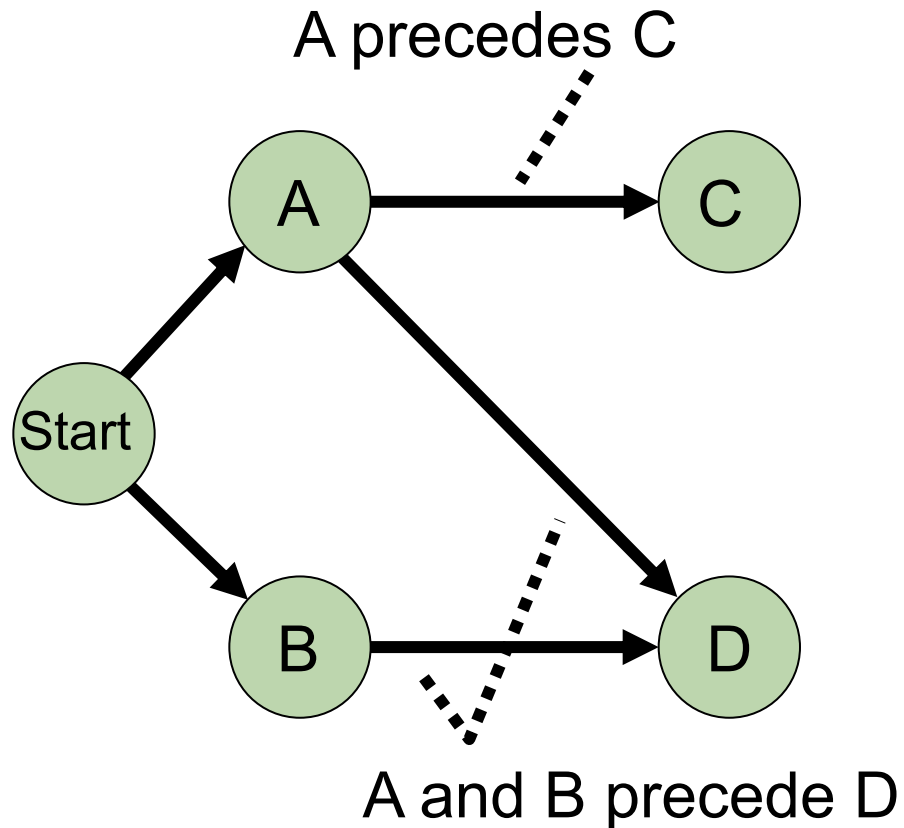
The plant needs to install an advanced computerized air pollution equipment within 16 weeks, to be ready for a major announcement.

8 activities need to be performed. Their precedence and inter-relationship as below:

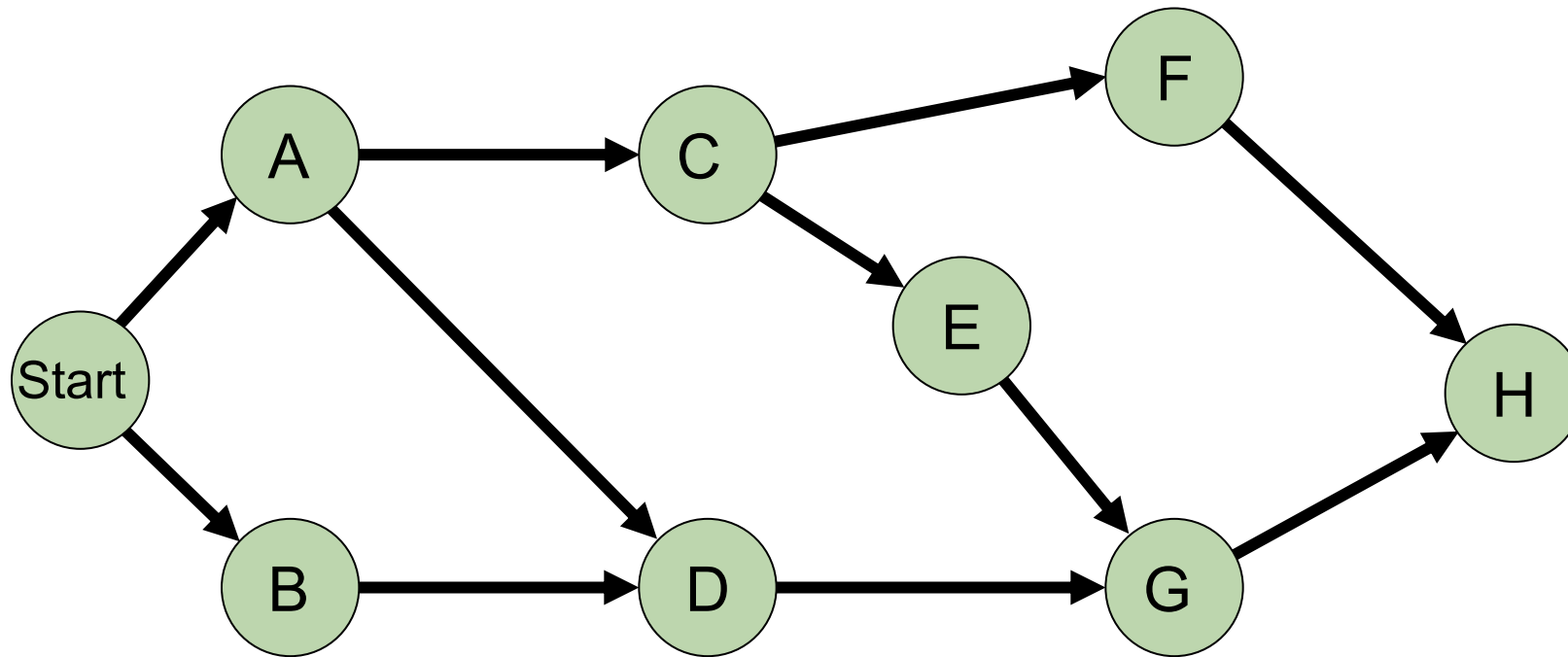
| Milwaukee Paper Manufacturing |                                    |                       |
|-------------------------------|------------------------------------|-----------------------|
| ACTIVITY                      | DESCRIPTION                        | Immediate predecessor |
| A                             | Build internal components          | —                     |
| B                             | Modify the roof and floor          | —                     |
| C                             | Construct a collection stack       | A                     |
| D                             | Pour concrete and install frame    | A, B                  |
| E                             | Build high temperature burner      | C                     |
| F                             | Install a pollution control system | C                     |
| G                             | Install air pollution device       | D, E                  |
| H                             | Inspect and test                   | F, G                  |



# Example: Milwaukee Paper: AON network



# Example: Milwaukee Paper: AON network



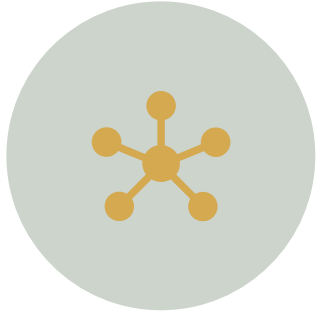
# Example: Milwaukee Paper: schedule

| Cycle time Milwaukee |                                    |              |
|----------------------|------------------------------------|--------------|
| ACTIVITY             | DESCRIPTION                        | Time (weeks) |
| A                    | Build internal components          | 2            |
| B                    | Modify the roof and floor          | 3            |
| C                    | Construct a collection stack       | 2            |
| D                    | Poor concrete and install frale    | 4            |
| E                    | Build high temperature burner      | 4            |
| F                    | Install a pollution control system | 3            |
| G                    | Install air pollution device       | 5            |
| H                    | Build internal components          | 2            |
|                      | Total if all in serie              | 25           |

Ch03, Heizer, Render et Munson 2019



# Example: Milwaukee Paper: critical path analysis



The critical path is the longest through the network.



The critical path is the shortest time for completing the project.



Any delay in CP impacts the whole project



No slack for critical path activities

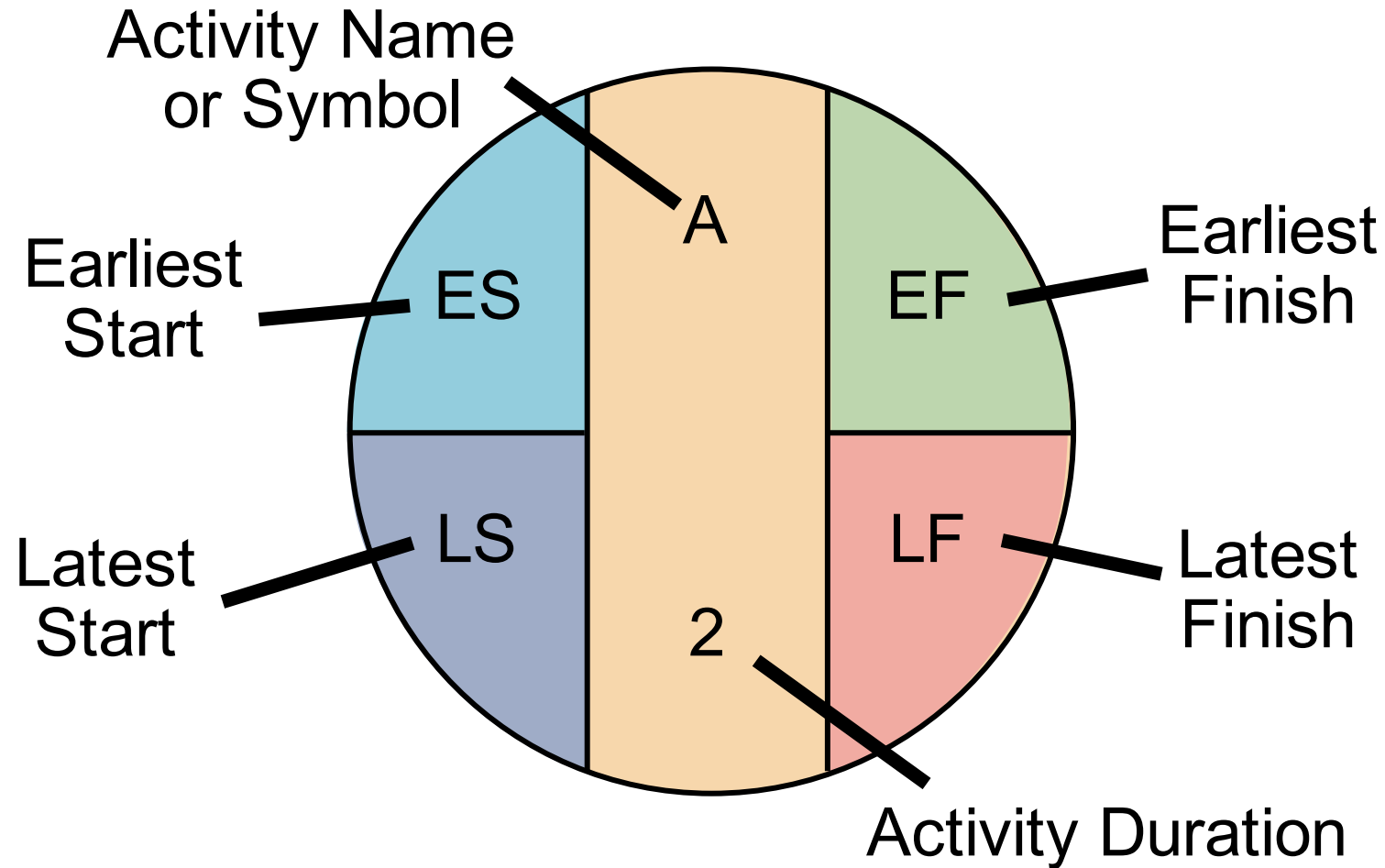
# Example: Milwaukee Paper:

## Critical path analysis terminology

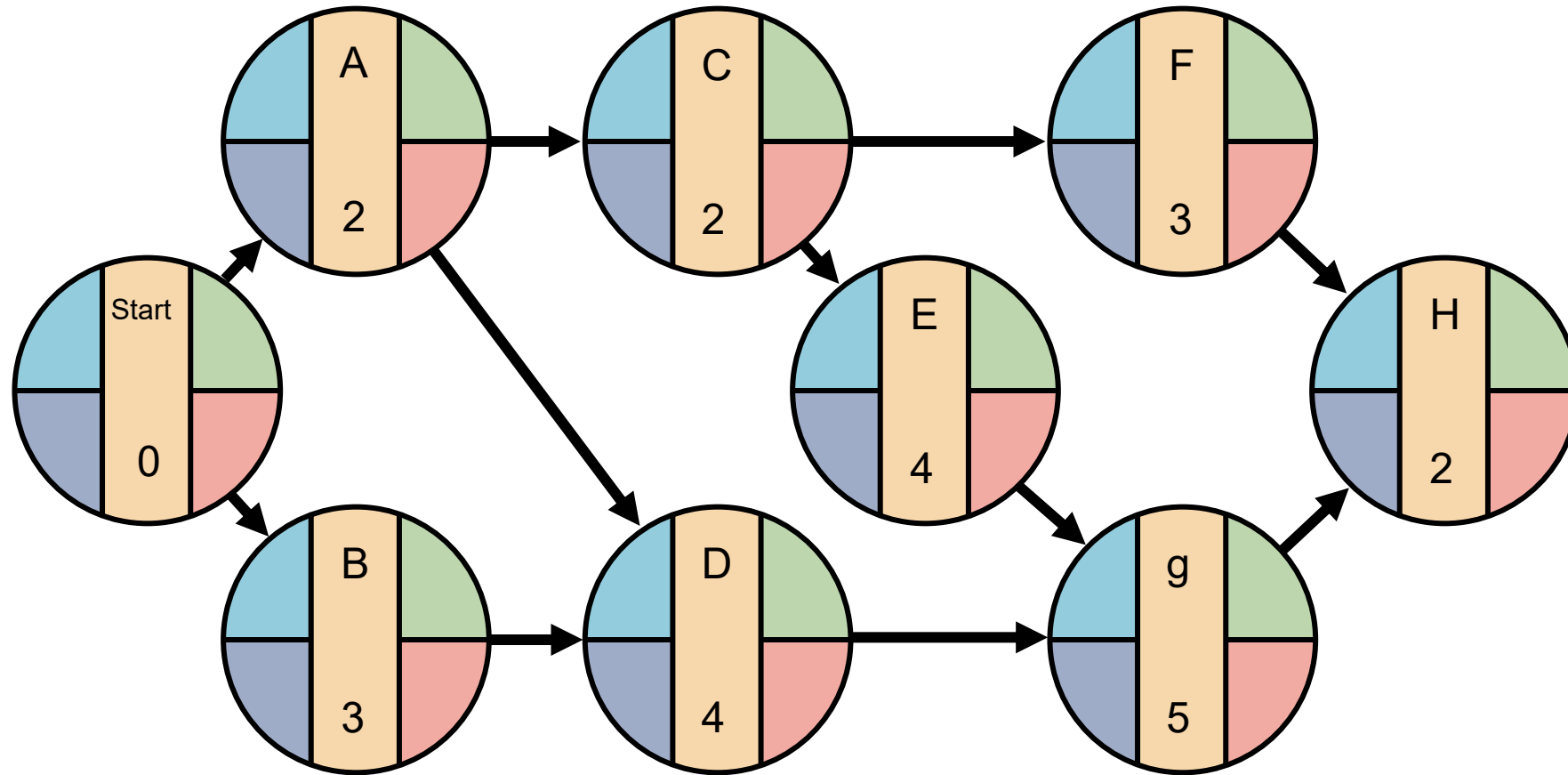
- **ES** = earliest time an activity can start assuming its predecessors are completed
- **EF** = earliest finished time of the activity
- Determined with a **forward path** analysis
- $ES = \text{Max}(\text{EF of all immediate predecessors})$
- $EF = ES + \text{activity time}$
- **LS** = Latest Start of an activity without impacting the project
- **LF** = Latest time an activity has to finish not to impact the total schedule
- Determined with a **backward path** analysis
- $LF = \min\{\text{LS of all immediate following activities}\}$
- $LS = LF - \text{activity time}$

Ch03, Heizer, Render et Munson 2019

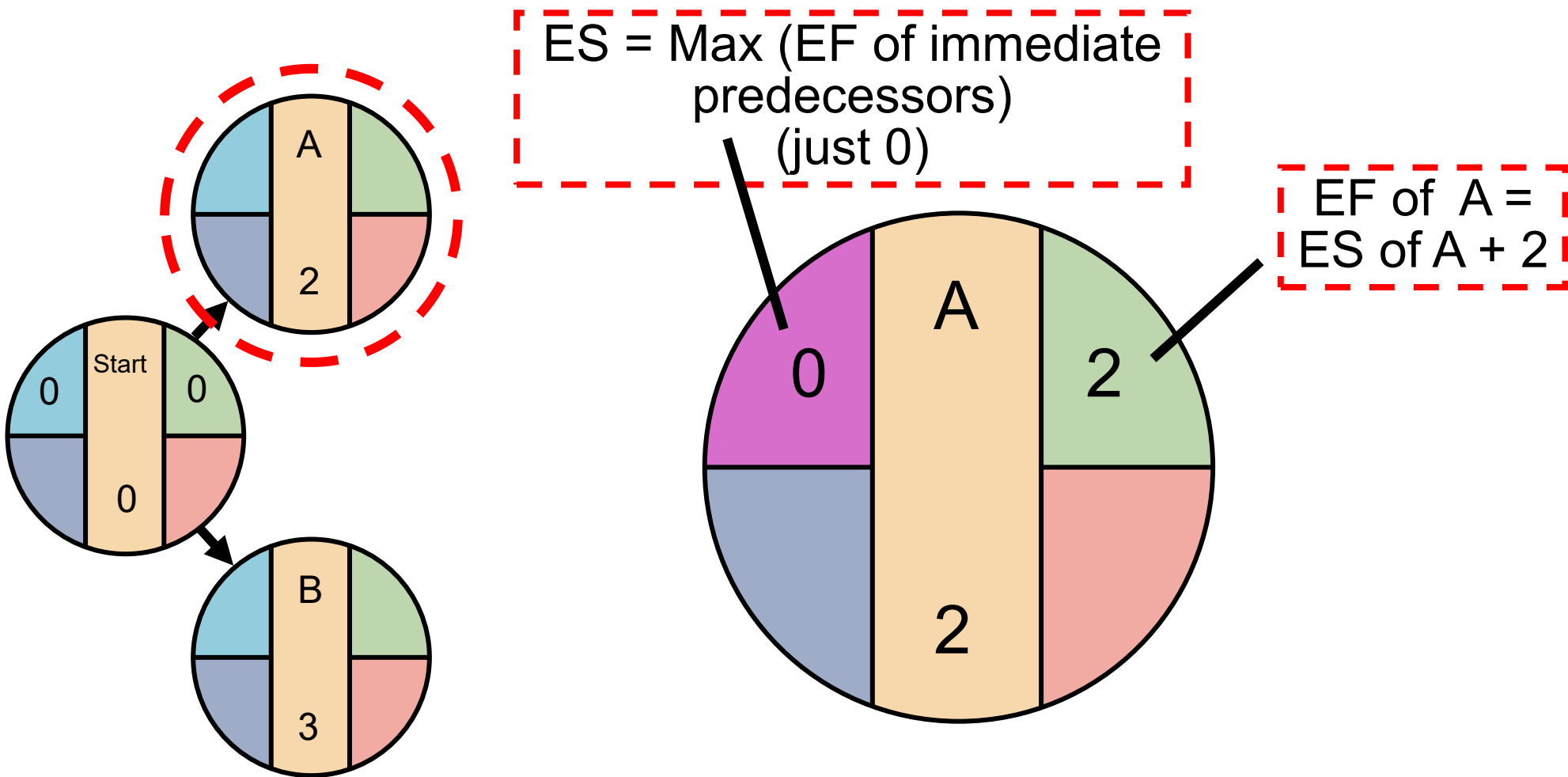
# Example: Milwaukee Paper: AON representation



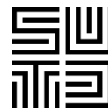
# Example: Milwaukee Paper: Forward path analysis



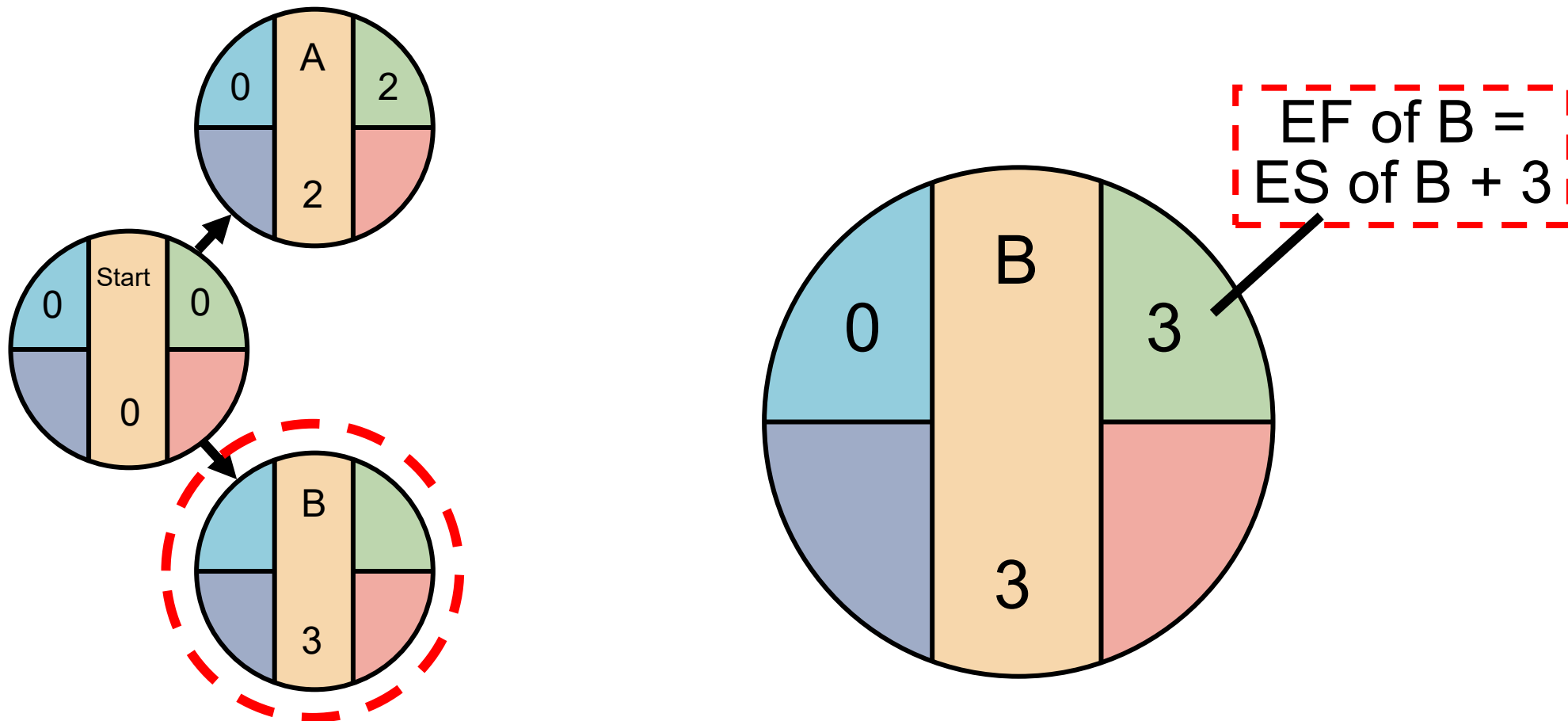
# Example: Milwaukee Paper: Forward path analysis





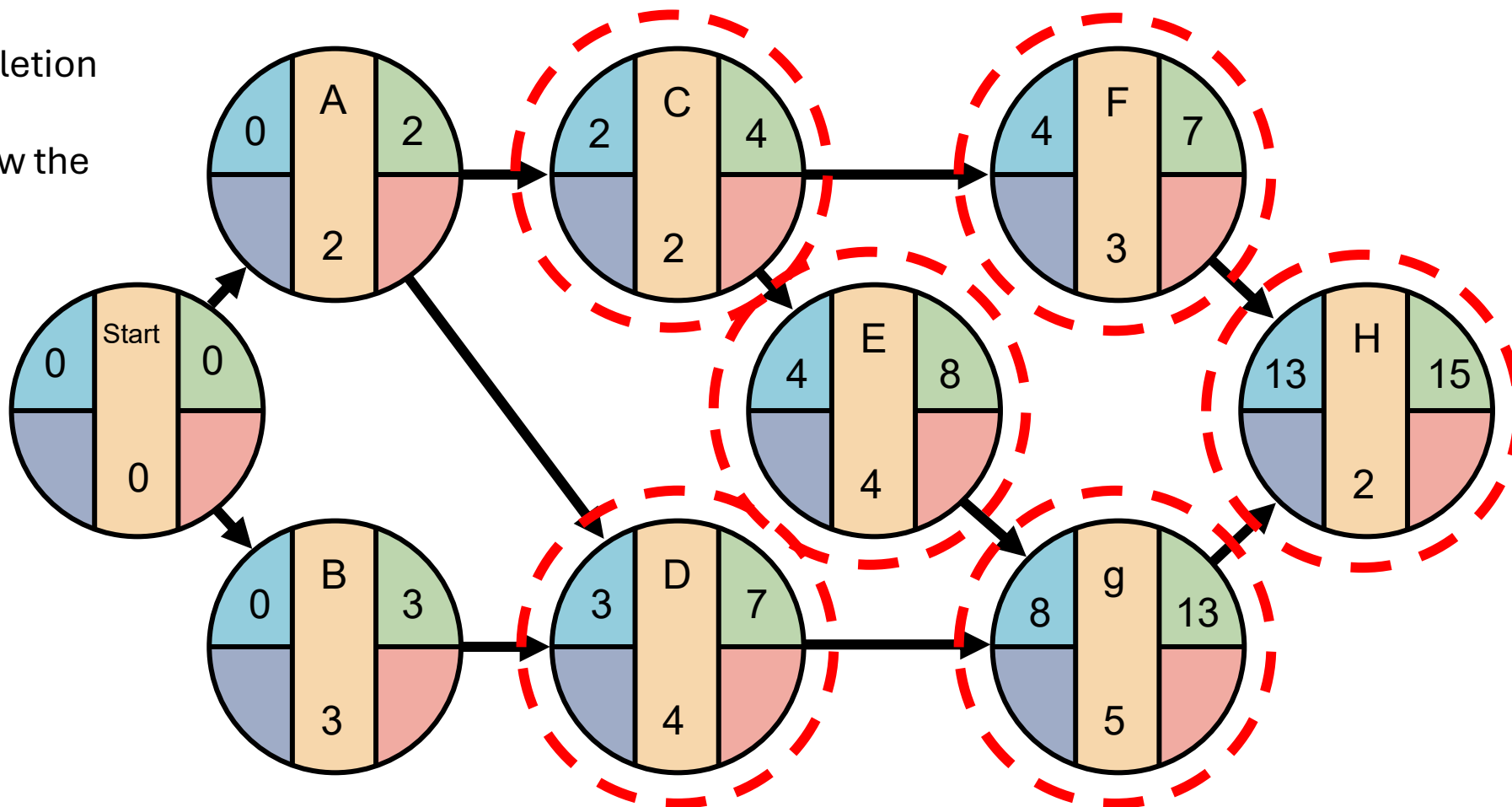


# Example: Milwaukee Paper: Forward path analysis

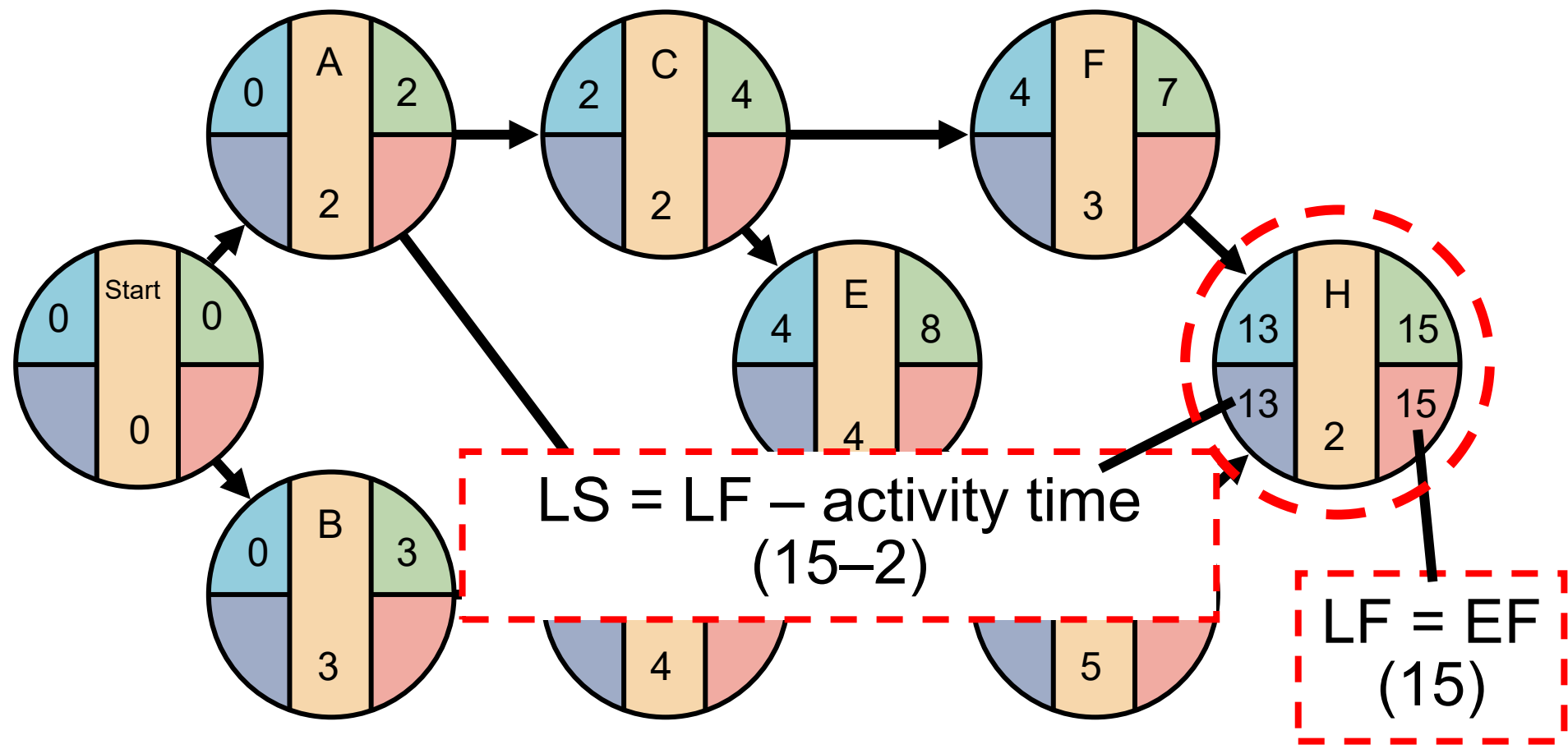


# Example: Milwaukee Paper: Forward path analysis

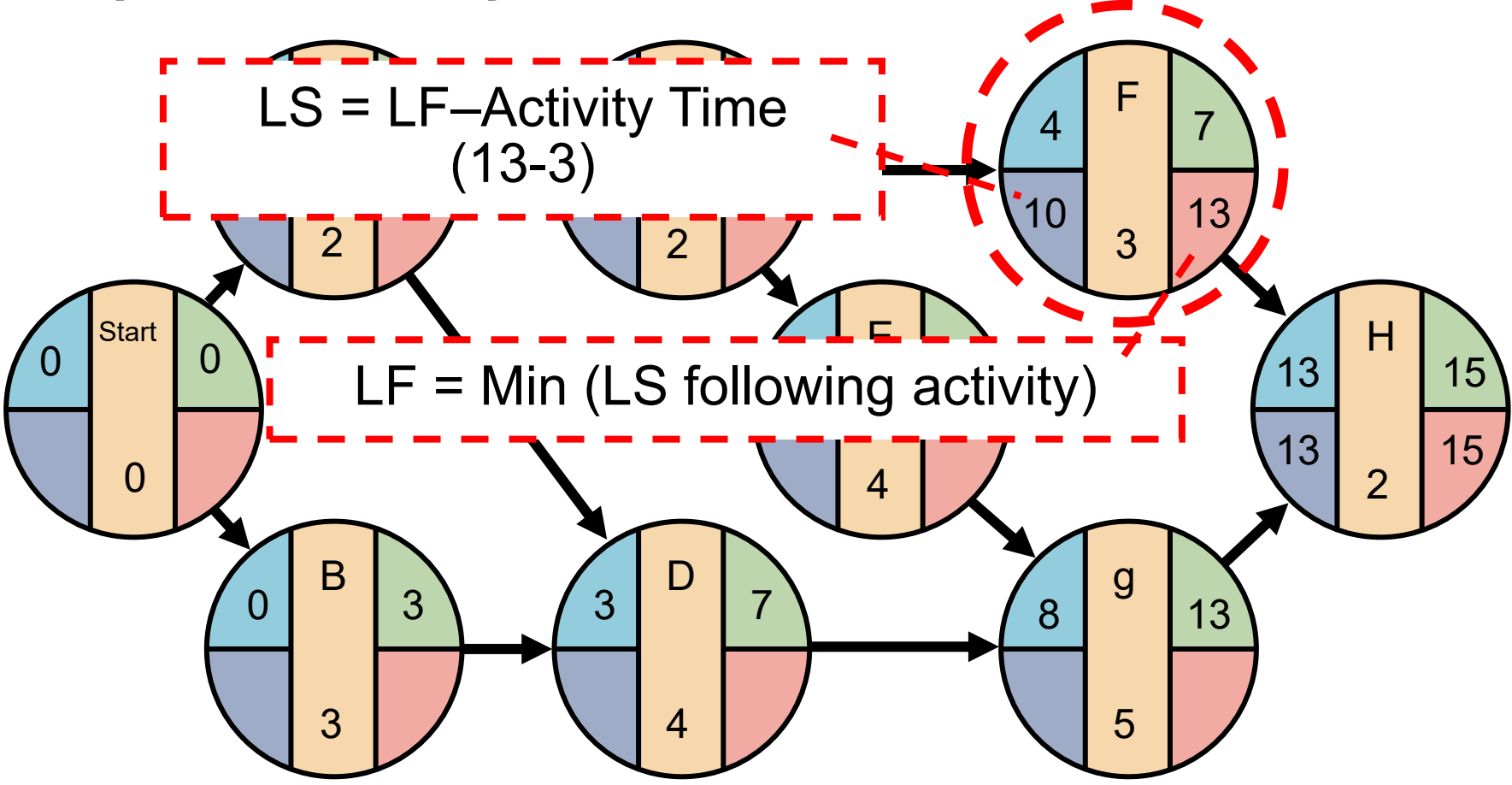
The earliest completion Time is 15 weeks.  
We do not yet know the Critical Path.



# Example: Milwaukee Paper: Backward path analysis

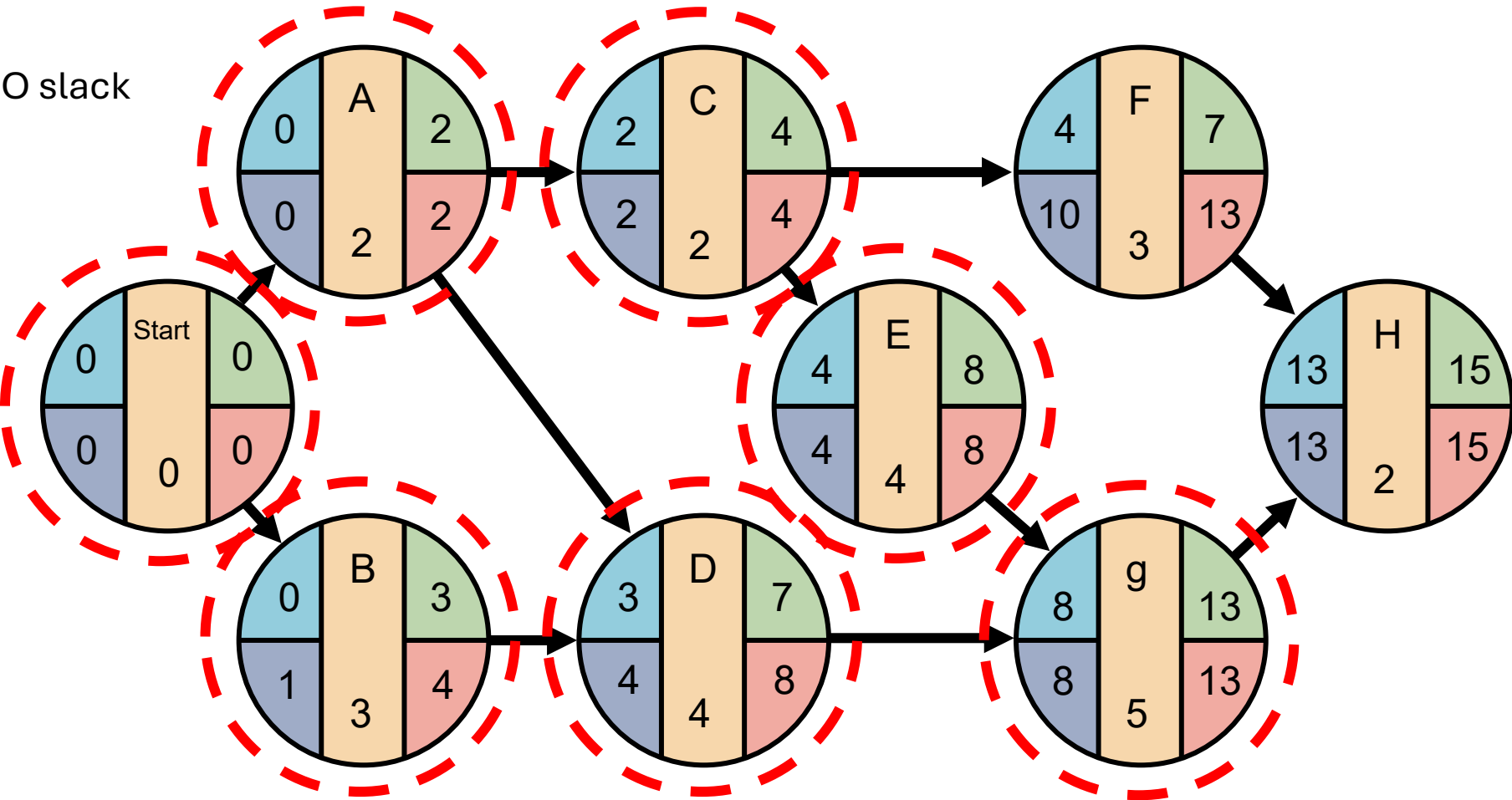


# Example: Milwaukee Paper: Backward path analysis



# Example: Milwaukee Paper: Backward path analysis

Critical Path:  
Activities with NO slack  
H-G-E-C-A





# Example: Milwaukee Paper:

## Activities with slack

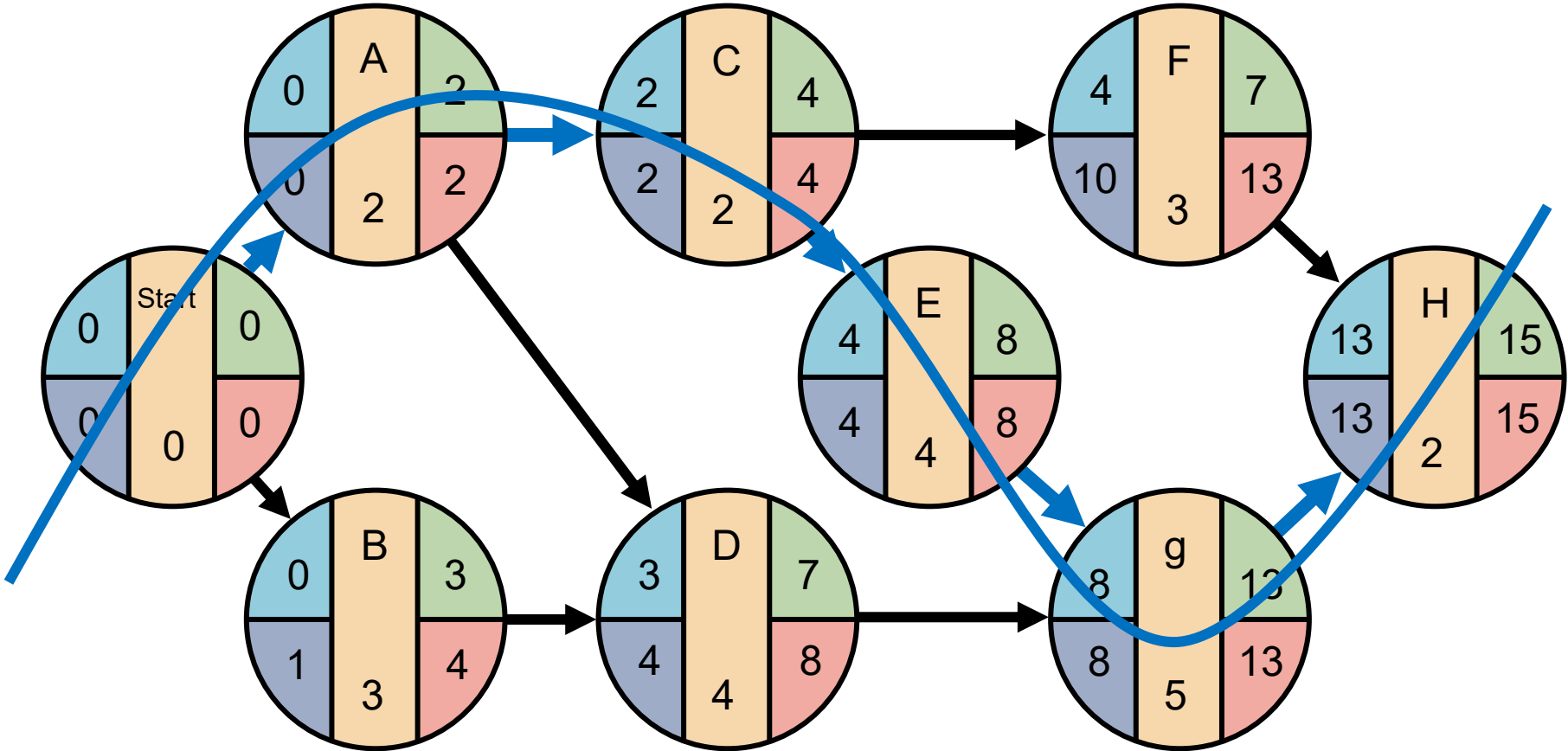
Slack means that the activity can be delayed without impacting the project.  
Slack happens when  $LF > EF$ .

In our example this happens  
for activities F, D and B.

| TABLE 3.3 Milwaukee Paper's Schedule and Slack Times |                         |                          |                       |                        |                  |                        |
|------------------------------------------------------|-------------------------|--------------------------|-----------------------|------------------------|------------------|------------------------|
| ACTIVITY                                             | EARLIEST<br>START<br>ES | EARLIEST<br>FINISH<br>EF | LATEST<br>START<br>LS | LATEST<br>FINISH<br>LF | SLACK<br>LS – ES | ON<br>CRITICAL<br>PATH |
| A                                                    | 0                       | 2                        | 0                     | 2                      | 0                | Yes                    |
| B                                                    | 0                       | 3                        | 1                     | 4                      | 1                | No                     |
| C                                                    | 2                       | 4                        | 2                     | 4                      | 0                | Yes                    |
| D                                                    | 3                       | 7                        | 4                     | 8                      | 1                | No                     |
| E                                                    | 4                       | 8                        | 4                     | 8                      | 0                | Yes                    |
| F                                                    | 4                       | 7                        | 10                    | 13                     | 6                | No                     |
| G                                                    | 8                       | 13                       | 8                     | 13                     | 0                | Yes                    |
| H                                                    | 13                      | 15                       | 13                    | 15                     | 0                | Yes                    |

Ch03, Heizer, Render et Munson 2019

# Example: Milwaukee Paper: Critical path



**Completion Time of 15 weeks**

# Exercise

| Activity | Time required (weeks) | Immediate Predecessor |
|----------|-----------------------|-----------------------|
| A        | 3                     | -                     |
| B        | 4                     | A                     |
| C        | 2                     | A                     |
| D        | 6                     | B,C                   |
| E        | 5                     | C                     |
| F        | 3                     | C                     |
| G        | 7                     | E                     |
| H        | 5                     | E, F                  |
| I        | 8                     | D, G, H               |

Draw the AON for this project

Find the critical path

Calculate the project duration



# Project Costing

We could try to improve the schedule by looking at expedite costing. The expediting cost function is assumed linear. As we work the expediting process it can be that a non critical activity becomes critical

| Activity | Time required (weeks) | Expedited time (weeks) | Normal cost | Expedite cost | Cost per week |
|----------|-----------------------|------------------------|-------------|---------------|---------------|
| A        | 3                     | 1                      | 1000        | 3000          | 1000          |
| B        | 4                     | 3                      | 4000        | 6000          | 2000          |
| C        | 2                     | 2                      | 2000        | 2000          |               |
| D        | 6                     | 4                      | 3000        | 6000          | 1500          |
| E        | 5                     | 4                      | 2500        | 3800          | 1300          |
| F        | 3                     | 2                      | 1500        | 3000          | 1500          |
| G        | 7                     | 4                      | 4500        | 8100          | 1200          |
| H        | 5                     | 4                      | 3000        | 3600          | 600           |
| I        | 8                     | 5                      | 8000        | 12800         | 1600          |

Cost per week=(expedited cost-normal cost)/(normal time – expedited time)

# Expediting a project (also called crashing)

- If we expedite all activities the total cost is 48300 versus the normal cost of 29500
- Let us assume we get a bonus of 1500 per week for reducing the project completion time
- If we rework our AON analysis we find that the completion time becomes 16weeks and we have 2 critical paths ACEGI and ACEHI

⇒ We pay 18800 and gain 13500

⇒ Not worth expediting everything

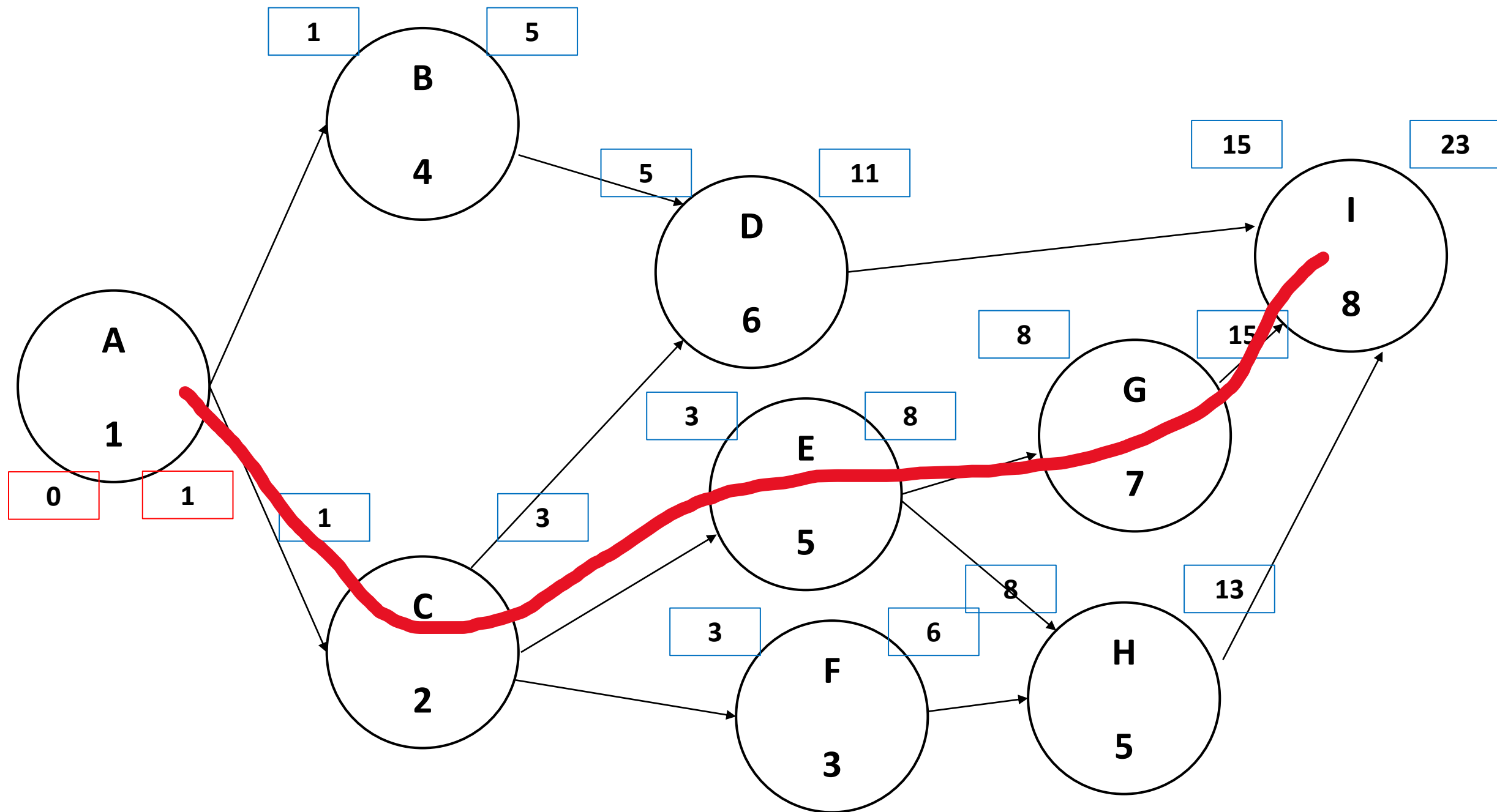
⇒ But there is likely an optimum

# Expediting a project (also called crashing)

- It is clear that we should focus on critical activities as they are driving our total completion time: A C E G I

| Activity | Time required (weeks) | Expedited time (weeks) | Cost per week |
|----------|-----------------------|------------------------|---------------|
| A        | 3                     | 1                      | 1000          |
| C        | 2                     | 2                      |               |
| E        | 5                     | 4                      | 1300          |
| G        | 7                     | 4                      | 1200          |
| I        | 8                     | 5                      | 1600          |

- Reducing A is the cheapest and does not introduce a new critical path and is profitable as we will pay 2000 but gain a 3000 bonus if we reduce to 1 week

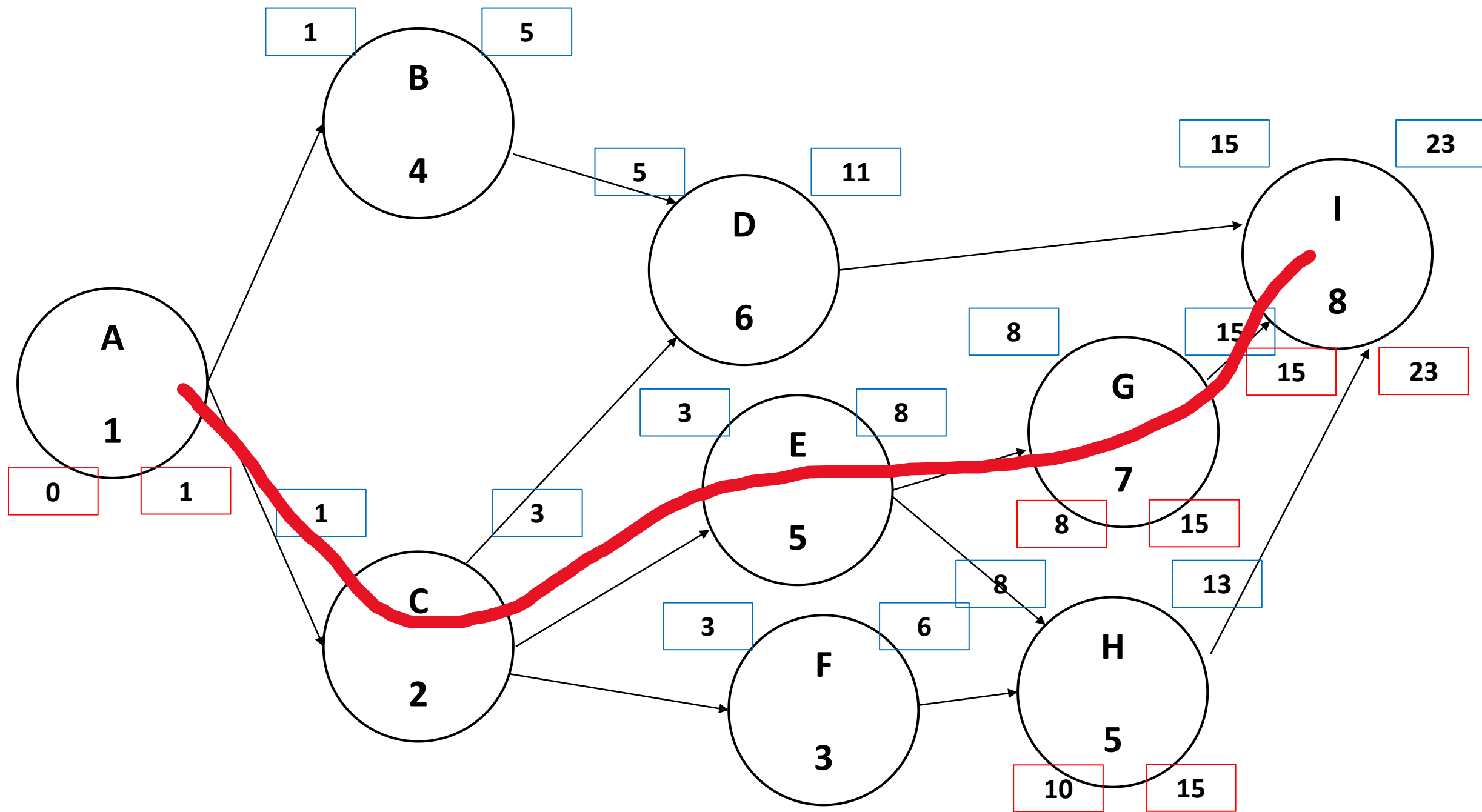


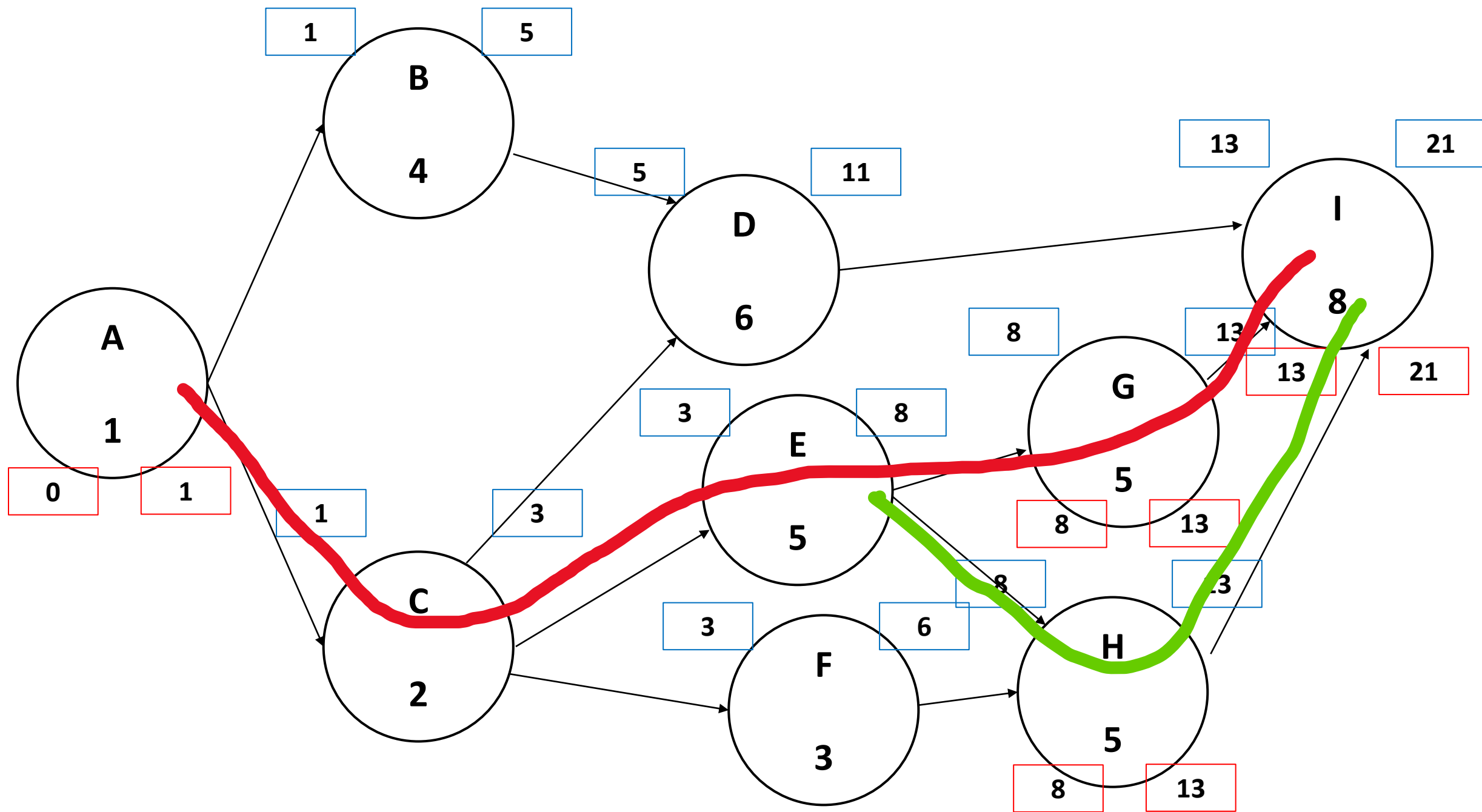
# Expediting a project (also called crashing)

- The next cheapest to consider is G. The critical path stays the same till G is reduced to 5 weeks. At that point we have 2 critical paths: ACEHI and ACEGI

| Activity | New Time required (weeks) | Expedited time (weeks) | Cost per week |
|----------|---------------------------|------------------------|---------------|
| A        | 1                         | 1                      |               |
| C        | 2                         | 2                      |               |
| E        | 5                         | 4                      | 1300          |
| G        | 5                         | 4                      | 1200          |
| I        | 8                         | 5                      | 1600          |
| H        | 5                         | 4                      | 600           |

- We have now a 21 weeks completion time
- If we want to further reduce we could select H as this is the least costly but we would also need to reduce G if we want an improvement and the other way around. This would cost 1800 and is thus not efficient.
- Thus we need to look at the other activities on the critical path. The only activity worth considering is E





# Expediting a project (also called crashing)

- Once E is reduced we have now a completion time of 20 weeks.
- Can we do more?

| Activity | New Time required (weeks) | Expedited time (weeks) | Cost per week |
|----------|---------------------------|------------------------|---------------|
| A        | 1                         | 1                      |               |
| C        | 2                         | 2                      |               |
| E        | 4                         | 4                      |               |
| G        | 5                         | 4                      | 1200          |
| I        | 8                         | 5                      | 1600          |
| H        | 5                         | 4                      | 600           |

- Since we cannot reduce G without reducing H there is no further economical solution
- The reduction costed us 5700 for a benefit of 7500





# What did we achieve so far?

- ✓ Identify project activities
- ✓ Draw a project network
- ✓ Compute ES, EF, LS and LF
- ✓ Identify a critical path
- ✓ Calculate a project completion time
- ✓ Crash a project by finding the expedite cost giving the best profit

We assumed activity times were fixed. This is the CPM approach.

# Project scheduling: PERT

We revise our assumptions and inject some randomness in activity times. This is the PERT approach.

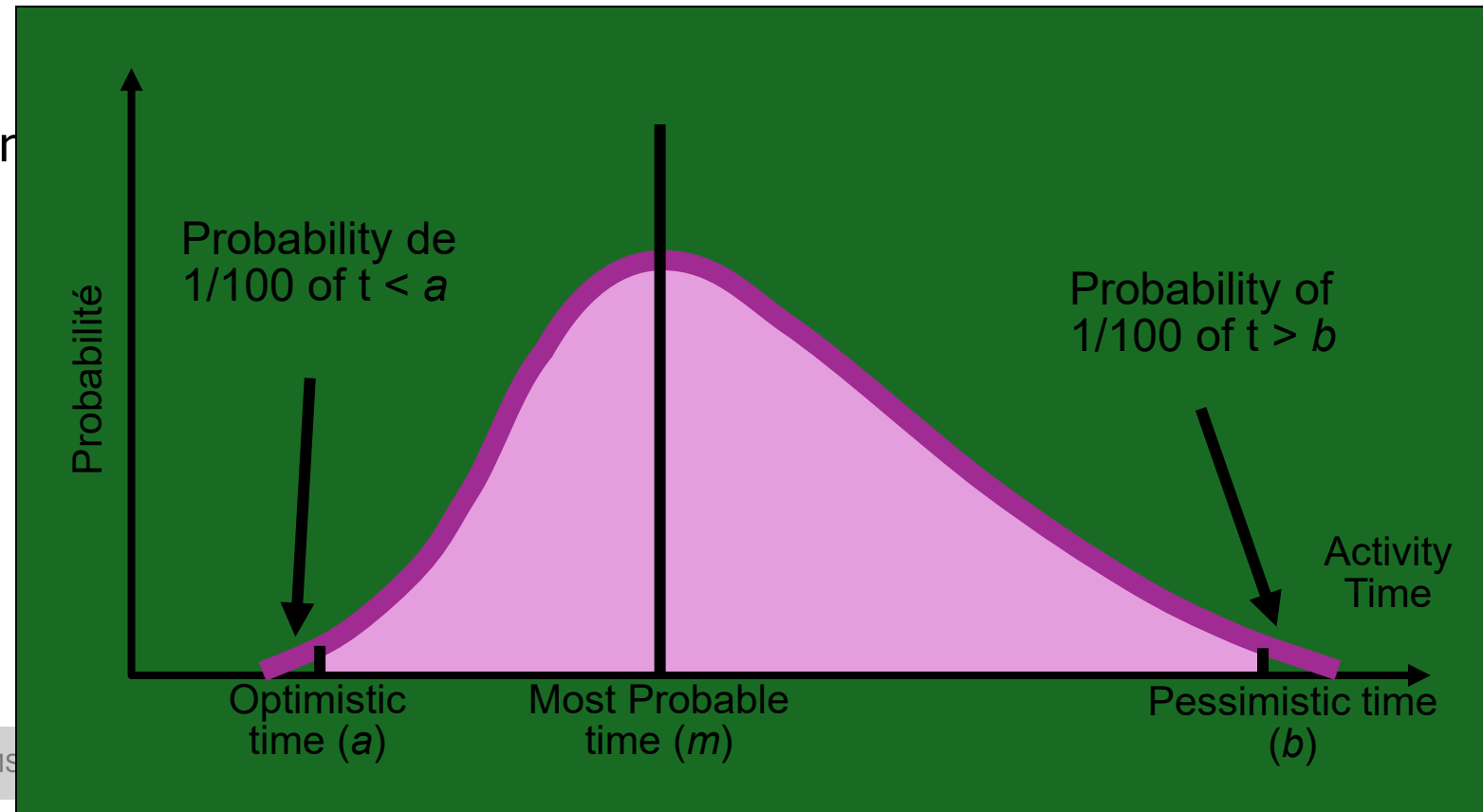
We use a probabilistic distribution of the activity time based on:

- Optimistic time  $a$ : small probability that the activity time will be  $< a$
- Pessimistic time  $b$ : small probability that the activity time will be  $> b$
- Most likely time  $c$

We assume a beta distribution for each activity time as our time is random within a finite interval.

We assume a normal distribution for the project completion time.

We assume all activity times are statistically independent.



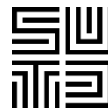
# Activity Time variability

Based on a beta distribution we have:

$$\text{Expected activity time, } t = \frac{a + 4m + b}{6}$$

$$\text{Variance (or dispersion) of activity completion time} = \left[ \frac{b - a}{6} \right]^2$$

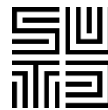
a: Optimisitic time  
m: most probable time  
B: Pessimistic time



# Applying PERT to Milwaukee Paper

| ACTIVITY | OPTIMIST<br>a | MOST<br>PROBABLE<br>m | PESSIMIST<br>b | Probable Time<br>$t=(a+4m+b)/6$ | VARIANCE<br>$=[(b-a)/6]^2$ |
|----------|---------------|-----------------------|----------------|---------------------------------|----------------------------|
| A        | 1             | 2                     | 3              |                                 |                            |
| B        | 2             | 3                     | 4              |                                 |                            |
| C        | 1             | 2                     | 3              |                                 |                            |
| D        | 2             | 4                     | 6              |                                 |                            |
| E        | 1             | 4                     | 7              |                                 |                            |
| F        | 1             | 2                     | 9              |                                 |                            |
| G        | 3             | 4                     | 11             |                                 |                            |
| H        | 1             | 2                     | 3              |                                 |                            |

Ch03, Heizer, Render et Munson 2019



# Applying PERT to Milwaukee Paper

| ACTIVITY                                                                                                                                                                                | OPTIMIST<br>a | MOST<br>PROBABLE<br>m | PESSIMIST<br>b | Probable Time<br>$t=(a+4m+b)/6$ | VARIANCE<br>$=[(b-a)/6]^2$ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|----------------|---------------------------------|----------------------------|
| A                                                                                                                                                                                       | 1             | 2                     | 3              | $= (1+4*2+3)/6 = 2$             | $= [(3-1)/6]^2 = 0,11$     |
| <b>NOTE:</b><br><br>By chance the probable times are the same as for the CPM analysis.<br><br>Thus the critical path will be the same as well as the total completion time of 15 weeks. |               |                       |                | $= (2+4*3+4)/6 = 3$             | $= [(4-2)/6]^2 = 0,11$     |
|                                                                                                                                                                                         |               |                       |                | $= (1+4*2+3)/6 = 2$             | $= [(3-1)/6]^2 = 0,11$     |
|                                                                                                                                                                                         |               |                       |                | $= (2+4*4+6)/6 = 4$             | $= [(6-2)/6]^2 = 0,44$     |
|                                                                                                                                                                                         |               |                       |                | $= (1+4*4+7)/6 = 4$             | $= [(7-1)/6]^2 = 1,00$     |
|                                                                                                                                                                                         |               |                       |                | $= (1+4*2+9)/6 = 3$             | $= [(9-1)/6]^2 = 1,78$     |
|                                                                                                                                                                                         |               |                       |                | $= (3+4*4+11)/6 = 5$            | $= [(11-3)/6]^2 = 1,78$    |
|                                                                                                                                                                                         |               |                       |                | $= (1+4*2+3)/6 = 2$             | $= [(3-1)/6]^2 = 0,11$     |



# Project Variance and std Dev for Milwaukee Paper

Based on our assumptions:

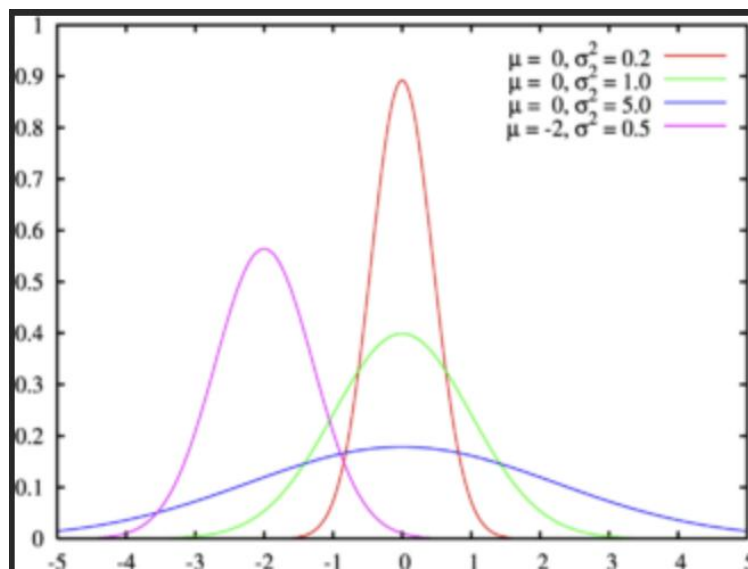
- $\sigma_p^2$  = Project variance =  $\sum$ (variances of activities on the critical path)
- $\sigma_p$  = Project standard deviation =  $\sqrt{\sigma_p^2}$
- Project variance = 3,11
- Project std dev = 1,76 weeks

A PERT assumption is that the project completion time follow a normal distribution.

# Statistics reminder

## Normal distribution vs standard normal distribution

- $N(\mu, \sigma)$  vs.  $N(0,1)$



To use the normal standard table we need to scale our own distribution by calculating the score  $z$ :

$$z = (X - \mu) / \sigma$$



[illegible]



| z    | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| -3.4 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0002 |
| -3.3 | 0.0005 | 0.0005 | 0.0005 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0003 |
| -3.2 | 0.0007 | 0.0007 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0005 | 0.0005 | 0.0005 |
| -3.1 | 0.0010 | 0.0009 | 0.0009 | 0.0009 | 0.0008 | 0.0008 | 0.0008 | 0.0008 | 0.0007 | 0.0007 |
| -3.0 | 0.0013 | 0.0013 | 0.0013 | 0.0012 | 0.0012 | 0.0011 | 0.0011 | 0.0011 | 0.0010 | 0.0010 |
| -2.9 | 0.0019 | 0.0018 | 0.0018 | 0.0017 | 0.0016 | 0.0016 | 0.0015 | 0.0015 | 0.0014 | 0.0014 |
| -2.8 | 0.0026 | 0.0025 | 0.0024 | 0.0023 | 0.0023 | 0.0022 | 0.0021 | 0.0021 | 0.0020 | 0.0019 |
| -2.7 | 0.0035 | 0.0034 | 0.0033 | 0.0032 | 0.0031 | 0.0030 | 0.0029 | 0.0028 | 0.0027 | 0.0026 |
| -2.6 | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |
| -2.5 | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| -2.4 | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| -2.3 | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| -2.2 | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| -2.1 | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| -2.0 | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| -1.9 | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| -1.8 | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| -1.7 | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| -1.6 | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| -1.5 | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| -1.4 | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| -1.3 | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| -1.2 | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| -1.1 | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| -1.0 | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| -0.9 | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| -0.8 | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| -0.7 | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| -0.6 | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| -0.5 | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| -0.4 | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| -0.3 | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |
| -0.2 | 0.4207 | 0.4168 | 0.4129 | 0.4090 | 0.4052 | 0.4013 | 0.3974 | 0.3936 | 0.3897 | 0.3859 |
| -0.1 | 0.4602 | 0.4562 | 0.4522 | 0.4483 | 0.4443 | 0.4404 | 0.4364 | 0.4325 | 0.4286 | 0.4247 |
| -0.0 | 0.5000 | 0.4960 | 0.4920 | 0.4880 | 0.4840 | 0.4801 | 0.4761 | 0.4721 | 0.4681 | 0.4641 |

# Reminder

Let us assume that the reaction times of a junior driver are normally distributed with a mean of 0,53 and a std dev of 0,11 sec.

1. What is the probability that a randomly selected young driver reacts faster than 0,65 sec?
2. What is the probability that a randomly selected young driver reacts between 0,4 and 0,6sec?
3. What is the probability that a randomly selected young driver has a reaction time larger than 0,8 sec?

# Milwaukee Paper

## Probability that the project is completed before the 16 weeks due date

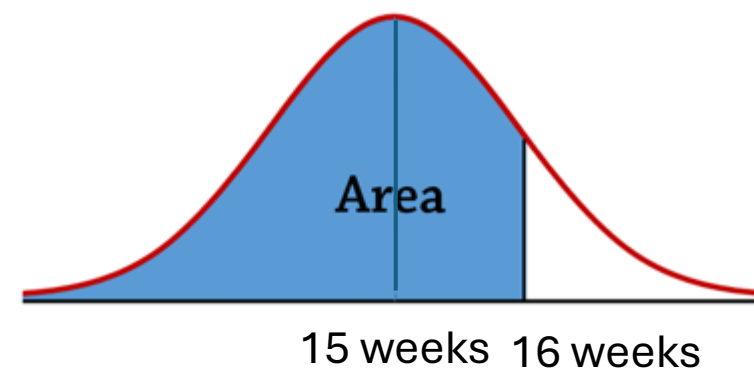
Based on our assumptions of a normal distribution we need to calculate the blue area.

1- Calculate the std dev between the completion date and the average:

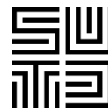
$$Z = \frac{(\text{Due Date} - \text{Expected Date of Completion})}{\sigma_p}$$

$$z=0,57$$

2- Refer to the std dev table to get the probability







# Milwaukee Paper

## Probability that the project is completed before the 16 weeks due date

The probability that the project is completed before week 16 is 71,57%

| z   | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |

# Milwaukee Paper

## Which due date gives them a 99% chance of on time completion?

- Look at the table for z value to have 99% chance or use xls (NORMSINV(0,99)); this is  $z=2.33$
- $$z = \frac{(\text{Due Date} - \text{Expected Date of Completion})}{\sigma_p}$$
- $z = 19.1$  weeks

# Conclusion

- Large projects are common in operations
- Using analytical techniques to manage project increases the chance of success
- The network representation helps to understand links between activities
- Monitoring progress is a critical pillar of a project

We have not covered risk assessment and mitigation but this is equally important to project success.

# THANK YOU

TRAILBLAZING A BETTER WORLD BY DESIGN.

